GEOSPATIAL ASSESSMENT OF FLOODING IMPACT AND VULNERABILITTY IN OGBARU L.G.A, ANAMBRA STATE (2018-2023)

ABSTRACT

Geospatial assessment of flood impact and vulnerability in Ogbaru Local Government Area (LGA), Anambra State, Nigeria, for the period 2018 to 2023. This study employes a comprehensive methodology integrating multi-temporal Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 optical imagery, Digital Elevation Models (DEMs), CHIRPS rainfall data, and socio-economic datasets within a Geographic Information System (GIS) and Google Earth Engine (GEE) framework, the research aimed to map flood extents, assess LULC changes and impacts, and generate a flood vulnerability map. Key findings reveal an average annual inundation of 2159 hectares, with peak extents in 2018 (2470.954 ha), demonstrating a strong positive correlation (r=0.65) with rainfall variability. Analysis indicated a cumulative 20% vegetation loss due to recurrent flooding and urban expansion, resulting in significant agricultural losses exceeding 750 hectares and substantial impacts on built-up areas (e.g., 300 ha in 2018). The generated Flood Vulnerability Index (FVI) map identified approximately 40% of Ogbaru LGA as highly vulnerable, encompassing about 60% of its population, primarily in low-elevation, densely populated riverine communities. The study concludes that geospatial technologies provide crucial insights for proactive disaster risk reduction. Recommendations include strengthening early warning systems, implementing integrated land use planning, promoting green infrastructure, and supporting resilient livelihoods to enhance community resilience in Ogbaru LGA.

Keywords: Flood Impact, Flood Vulnerability, Remote Sensing, GIS, Ogbaru LGA, Land Use/Land Cover, Sentinel, Nigeria.

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CHAPTER ONE

INTRODUCTION

1.0 INTRODUCTION TO STUDY

In recent decades, flooding has emerged as one of the most pervasive and devastating natural hazards globally. The widespread and recurring nature of these events exerts immense socio-economic disruption, environmental degradation, and a significant loss of human lives. This reality is consistently highlighted in major international reports, such as those from the United Nations

Office for Disaster Risk Reduction (UNDRR) and the World Bank, which identify floods as a leading cause of disaster-related economic losses and displacement. The increasing frequency and intensity of these hazards, exacerbated by climate change and rapid urbanization, underscore the urgent need for effective flood risk management strategies, particularly in vulnerable regions. (UNDRR, 2019; World Bank, 2019). In an era marked by accelerating climate change (IPCC, 2023), unprecedented rates of urbanization (UN-Habitat, 2020), and often inadequate land use planning (World Bank, 2019), the frequency, intensity, and spatial extent of flood events have witnessed a notable increase (UNDRR, 2021). These dynamics pose formidable challenges to human settlements, agricultural systems, and overall infrastructural resilience, particularly within low-lying coastal and expansive riverine areas (Adewuyi & Ologunorisa, 2007). The African continent, given its physiographic characteristics, socio-economic vulnerabilities, and climatic patterns, is disproportionately affected by these phenomena, with recurrent flood episodes impeding sustainable development and exacerbating poverty (AfDB, 2022).

Nigeria, strategically positioned within a tropical climate zone and characterized by extensive river networks, experiences annual flood occurrences of varying magnitudes (Nkwunonwo et al., 2014). Its riverine communities, especially those nestled along the corridors of major water bodies like the River Niger and its tributaries, are acutely susceptible to inundation (Ologunorisa & Adeyemo, 2007). Anambra State, located in the south eastern geopolitical zone of Nigeria, is no exception; it is frequently impacted by the seasonal overflow of the River Niger, leading to widespread and often catastrophic flooding (Eze & Eze, 2019). Among its constituent Local Government Areas (LGAs), Ogbaru stands out as a particularly vulnerable area, owing to its direct proximity to the River Niger and its heavy reliance on flood-sensitive economic activities such as agriculture and fishing (NEMA, 2023).

This study undertakes a comprehensive geospatial assessment of flood impact and vulnerability within Ogbaru LGA over a critical five-year period, from 2018 to 2023 (Current Study Period, 2024). By meticulously leveraging the capabilities of Geographic Information System (GIS) and Remote Sensing techniques, this research seeks to dissect and analyze the intricate spatial extent, temporal frequency, and severity of flood events that have besieged the area (Chapman & Smith, 2018). Furthermore, it aims to systematically evaluate the multifaceted socio-economic and environmental impacts stemming from these floods, and critically, to accurately identify and map the areas and populations most vulnerable to future inundations (Ajayi et al., 2020). The integration of advanced geospatial technologies, specifically Synthetic Aperture Radar (SAR) and optical satellite imagery, offers an inherently robust and scientifically rigorous framework for understanding complex flood dynamics, quantifying damage, and ultimately informing the formulation of precise, evidence-based disaster risk reduction strategies (Richards & Jia, 2006). This approach moves beyond anecdotal observations to provide a granular, data-driven understanding crucial for building resilient communities in the face of escalating flood threats (UNISDR, 2015).

1.1 BACKGROUND TO THE STUDY

Floods, in their most fundamental definition, represent a natural phenomenon where an area of land, typically dry, becomes submerged by water (Kundzewicz & Plate, 2011). These events are triggered by a confluence of factors, including prolonged heavy rainfall, the overflow of rivers and lakes, coastal storm surges, dam failures, and inadequate drainage systems (Ward et al., 2020). While inherent to many natural hydrological cycles, the anthropogenic alteration of landscapes and the profound effects of climate change have undeniably intensified the global impact of flooding in recent decades (Hirabayashi et al., 2013). This intensification has led to a global imperative for the development and implementation of robust mitigation, adaptation, and preparedness strategies (Global Commission on Adaptation, 2019). Developing nations, frequently characterized by burgeoning population densities, the proliferation of informal settlements within hazard-prone floodplains, and often limited adaptive capacities due to underdeveloped infrastructure and resource constraints, disproportionately bear the brunt of these escalating disasters (Dilley et al., 2005).

The global landscape of flood disasters reveals a clear and accelerating upward trend in both economic losses and human displacement. According to data compiled by the United Nations Office for Disaster Risk Reduction (UNDRR) and the EM-DAT database, floods accounted for over 40% of all natural disasters between 2000 and 2019, impacting billions of people and causing trillions of dollars in cumulative damages (UNDRR, 2021; EM-DAT, 2022). This escalating trend is closely linked to the compounding effects of accelerating climate change (IPCC, 2023) and unprecedented rates of urbanization, which often push populations into flood-prone areas (UN-

Habitat, 2020). The resulting challenges to human settlements, agricultural systems, and overall infrastructural resilience are particularly pronounced within low-lying coastal and expansive riverine areas (Adewuyi & Ologunorisa, 2007).

This global pattern finds a strong and particularly resonant echo in the African context, where vulnerability to climate-related hazards, including floods, is exceptionally high. This is due to a complex interplay of factors, including often-weak governance structures, widespread poverty, and a heavy reliance on climate-sensitive livelihoods like subsistence agriculture (African Climate Policy Centre, 2012). The consequences of this heightened vulnerability are severe and cyclical. Recurrent flood episodes impede sustainable development, damage critical infrastructure, and exacerbate existing poverty levels, posing formidable obstacles to the continent's long-term economic and social progress (AfDB, 2022).

Nigeria, a West African nation endowed with an extensive network of rivers, including the mighty River Niger and its principal tributary, the River Benue, is acutely susceptible to annual flooding (Oruonye & Musa, 2013). The country's geographical position, coupled with its distinct bimodal rainfall pattern, ensures that a significant portion of its landmass experiences inundation, particularly during the lengthy wet season, which typically spans from June to October (Federal Ministry of Environment, 2012). Historical flood events serve as stark reminders of the nation's perennial vulnerability. The devastating nationwide floods of 2012, for instance, were unprecedented in their scale, submerging vast swathes of land across 30 of Nigeria's 36 states, displacing millions, and causing estimated damages running into billions of Naira (NEMA, 2012; ActionAid, 2013). Subsequent recurrent occurrences, including significant events in 2018, 2020, and the exceptionally severe 2022 floods, have further underscored the nation's persistent susceptibility and the urgent, unaddressed need for comprehensive and proactive flood risk management strategies (NEMA Annual Reports, 2018-2022). These events have not only resulted in widespread displacement and the destruction of property and critical infrastructure but have also led to substantial losses of agricultural yields, severe disruption of economic activities, and alarming outbreaks of waterborne diseases, collectively undermining national development efforts (IFRC, 2022).

1.2 STUDY AREA

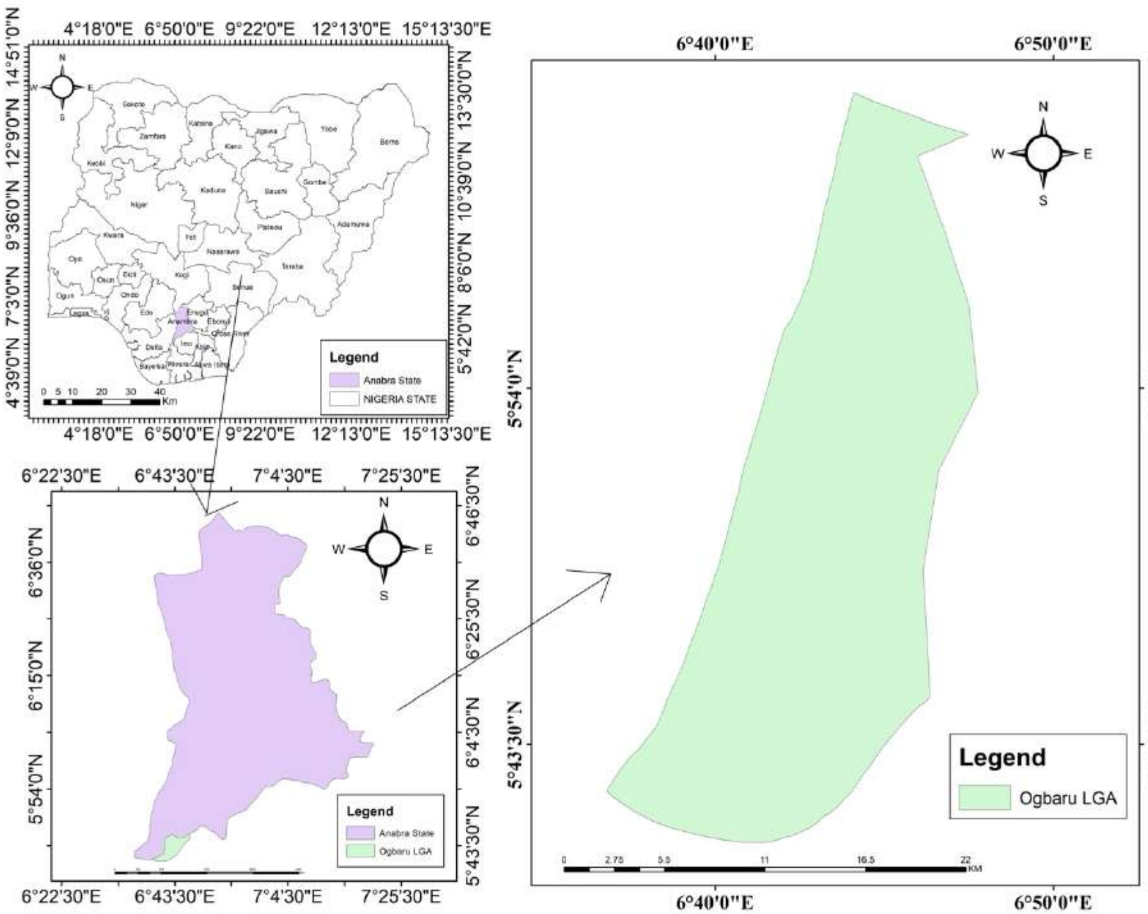
Ogbaru LGA, situated in Anambra State, southeastern Nigeria, is a riverine territory highly susceptible to annual flooding due to its proximity to major waterways. Geographically, it lies between latitudes 5°50'N and 6°05'N and longitudes 6°40'E and 6°50'E, bordered by the River Niger to the west (from Okpoko town to Ogwu-ikpele boundary with Rivers State) and the Orashi River to the east (along Ogwu-aniocha and Osomari forest reserve). To the northeast, it adjoins Ihiala, Okija, Owerri, Onitsha road, Ozubulu, Oraifite, and Oba.

Physical and Human Geography Features:

- **Land Area:** Approximately 387.7 km² (149.7 sq mi).
- **Topography:** Predominantly flat, low-lying coastal and riverine plains with elevations averaging 25 m above sea level, dominated by shallow aquifers and floodplains that exacerbate inundation risks.
- **Vegetation:** Comprises wetlands, mangroves, degraded lowland forests, and extensive agricultural lands, including farmlands prone to submersion.
- **Climate:** Tropical monsoon climate characterized by high humidity, mean annual temperatures of 27–32°C, and bimodal rainfall peaking in July–September. Annual precipitation ranges from 1,800–2,200 mm, with a lengthy rainy season from March to October (9 months) and a short dry season, contributing to frequent flooding.
- **Population:** Estimated at 318,200 (2022), primarily Igbo ethnic group, with high density in semi-urban settlements like Atani, Odekpe, and Ossomala. Over 60% reside in flood-vulnerable areas, relying on subsistence agriculture, fishing, and trade.
- **Economy and Infrastructure:** Features a Nigerian naval base, industrial river harbor, refinery, and ongoing federal road constructions linking to Rivers State and the Second Niger Bridge. Agriculture dominates, with crops like rice, yam, and cassava, but recurrent floods disrupt livelihoods.
- **Flood History:** Ogbaru experiences chronic flooding due to the shallow River Niger depth and overflow during heavy rains. Notable events include: 2018 floods killing 12 and polluting rivers; 2020 displacing 1,000 and damaging farms; 2022 submerging homes,

farmlands, and markets, with 76 drowning in a boat incident and 286,000 affected per NEMA reports.

This ecological and socio-economic profile underscores Ogbaru's vulnerability, making it a critical site for flood research to inform resilience-building.



Despite the recurrent and well-documented nature of these catastrophic flood disasters, traditional methods of data collection and impact assessment in areas like Ogbaru have often proven insufficient (UN-SPIDER, 2016). Such methods, frequently relying on manual ground surveys or generalized administrative reports, fall short in providing the precise spatial detail, comprehensive coverage, and dynamic temporal insights necessary for effective pre-disaster planning and post-disaster recovery (Adeoye et al., 2018). This critical data gap underscores the indispensable and transformative role that advanced geospatial technologies, specifically Remote Sensing and Geographic Information Systems (GIS), can play in augmenting our understanding of flood dynamics and supporting evidence-based decision-making processes (Longley et al., 2011). Remote Sensing, through its ability to acquire synoptic data on land cover, water extent, and precise topographic features from satellite platforms, offers unparalleled capabilities for rapid and accurate mapping of inundated areas (Wang et al., 2019). Concurrently, GIS provides the analytical framework for the integration, manipulation, and visualization of diverse spatial data layers, enabling the precise delineation of flood-prone areas, the identification of affected assets, and the assessment of vulnerable elements within the landscape (Burrough & McDonnell, 2015). The synergy between these technologies provides a powerful suite of tools for addressing the complexities of flood risk in vulnerable regions like Ogbaru LGA (Dewan, 2013).

1.3 STATEMENT OF THE PROBLEM

Ogbaru Local Government Area has, over several decades, consistently been identified and confirmed as one of the most chronically flood-prone LGAs within Anambra State, enduring annual flood events of varying magnitudes, particularly during the peak of Nigeria's rainy season (NEMA, 2023; Eze et al., 2019). While these floods are, to some extent, a recurrent environmental phenomenon intrinsically linked to its geographical position within the River Niger floodplain, their impacts have demonstrably intensified and become increasingly severe throughout the period spanning 2018 to 2023 (Adanu & Onwuemena, 2021). This escalating severity has translated into profound and quantifiable losses, encompassing human lives, critical livelihoods, invaluable infrastructure, and vast expanses of prime agricultural lands (UNDRR, 2021; World Bank, 2020). Communities nestled within Ogbaru LGA face a perennial cycle of displacement, destruction of residential structures, irreversible damage to productive farmlands, significant disruption of essential transportation networks, and an alarming increase in the prevalence of waterborne diseases following each flood episode (Okeke & Obike, 2018; IFRC, 2022).

Chapter Four of this study provides empirical evidence of these escalating impacts. For instance, the analysis reveals that flood extents averaged approximately 2159 hectares annually across the study period, with a significant peak in 2018 at 2470.954 hectares, representing about 5.6% of

Ogbaru's total land area. The impact on land use/land cover (LULC) has been substantial, with post-flood analyses indicating notable vegetation declines, culminating in a cumulative loss of approximately 20% across the five-year period. This conversion is often to water or bareland, indicative of inundation and erosion (Adanu & Onwuemena, 2021). Furthermore, the agricultural sector, the bedrock of Ogbaru's economy, has suffered immensely, with total agricultural impacts (proxy by flooded vegetation/forest) exceeding 750 hectares over the study period, with 2018 experiencing the highest losses at 460.678 hectares. Built-up areas have also been significantly affected, with over 300 hectares of built-up land flooded in 2018 alone, demonstrating rising impacts on urban settlements and infrastructure in subsequent years (e.g., 2022), signalling an alarming trend of urban encroachment into inherently flood-prone zones (Ologunorisa & Adeyemo, 2007)

Despite the palpable and well-documented recurring nature of these flood disasters and the quantifiable losses, there remains a critical and persistent deficit of precise, up-to-date, and spatially explicit information (UN-SPIDER, 2016). This lack of data pertains specifically to the actual extent of inundation at a granular level, the precise identification of specific areas and assets most severely affected, and a comprehensive understanding of the socio-economic vulnerability profiles of different communities within Ogbaru LGA (Adeoye et al., 2018). Existing assessments often rely on generalized statistics, broad administrative boundaries, or anecdotal accounts, failing to capture the dynamic and nuanced nature of flood events or to comprehensively map the intricate underlying factors that contribute to differential vulnerability across the landscape (Ward et al., 2020).

This pervasive absence of detailed geospatial information creates significant impediments to effective pre-disaster planning, the establishment and operationalization of robust early warning systems, the efficiency of post-disaster relief and recovery efforts, and the strategic implementation of sustainable long-term flood mitigation measures (UNISDR, 2015). Without a clear, scientifically derived understanding of precisely where floods occur, who is most affected by their direct and indirect consequences, and crucially, *why* certain areas and populations exhibit higher levels of susceptibility, response efforts risk being inefficient, critical resources may be misallocated, and efforts towards building long-term community resilience are significantly hampered (Dilley et al., 2005). Therefore, there is an urgent and critical need for a rigorous geospatial assessment that not only quantifies the direct and indirect flood impact but also systematically identifies and maps areas of elevated vulnerability in Ogbaru LGA for the period between 2018 and 2023. Such a study will provide an indispensable, scientifically defensible basis for proactive disaster risk reduction, informing policy, guiding interventions, and ultimately fostering greater resilience within the affected communities (Adelekan, 2011).

1.3 AIM AND OBJECTIVES OF THE STUDY

The overarching **aim** of this study is to conduct a comprehensive geospatial assessment of flood impact and vulnerability in Ogbaru Local Government Area, Anambra State, specifically covering the period from 2018 to 2023, by rigorously employing Remote Sensing and Geographic Information System (GIS) techniques.

This study is guided by the following specific objectives:

i. map the spatial extent and frequency of flood events in Ogbaru LGA between 2018 and 2023;
ii. assess the impact of floods on land cover, infrastructure, and agricultural areas within the study period;
iii. identify and map socio-economic factors contributing to flood vulnerability in Ogbaru LGA; and
iv. generate a comprehensive flood vulnerability map for Ogbaru LG.

1.4 Research Questions

In pursuit of the stated aim and objectives, this study will seek to provide answers to the following fundamental research questions:

i. What is the precise spatial extent and temporal frequency of flood events observed in Ogbaru LGA during the period of 2018 to 2023?
ii. To what quantitative extent have floods impacted different land cover types, critical infrastructure, and agricultural lands within Ogbaru LGA over the study periodWhat are

the predominant socio-economic factors that significantly contribute to the varying degrees of flood vulnerability across Ogbaru LGA?

iii. Based on the integrated geospatial analysis, which specific areas within Ogbaru LGA are characterized by the highest levels of vulnerability to flood hazards?

iv. What scientifically informed and spatially explicit flood risk management strategies can be recommended for Ogbaru LGA to enhance its resilience to future flood events, drawing directly from the findings of this geospatial assessment?

1.5 SIGNIFICANCE OF THE STUDY

The present study holds profound and multi-faceted significance for a diverse range of stakeholders within Ogbaru LGA and extends its relevance to broader disaster management frameworks in Nigeria (UNDRR, 2021). Firstly, it addresses a critical knowledge gap by providing a comprehensive, precise, and spatially explicit understanding of flood impact and vulnerability within the study area (Ajayi et al., 2020). This is particularly crucial given the historical reliance on generalized data, which often hinders targeted interventions (UN-SPIDER, 2016). The meticulously generated flood extent maps, land use/land cover change maps, and the derived flood vulnerability map will serve as indispensable tools for various government and non-governmental entities (Burrough & McDonnell, 2015). These include local government agencies within Ogbaru LGA, national and state emergency management organizations such as the National Emergency Management Agency (NEMA) and the State Emergency Management Agency (SEMA) in Anambra, as well as non-governmental organizations (NGOs) actively engaged in humanitarian aid, community development, and environmental conservation initiatives (NEMA, 2023; IFRC, 2022). These maps will enable these bodies to make more informed decisions regarding resource allocation, emergency response, and long-term planning (Adelekan, 2011).

Secondly, the empirical findings and spatial intelligence yielded by this research will directly inform and facilitate the development and implementation of more targeted, efficient, and proactive flood mitigation and adaptation strategies (UNISDR, 2015). By precisely identifying specific areas and populations most at risk, resources can be strategically channeled towards resilient infrastructure planning (e.g., elevating structures, designing flood-resistant buildings), the establishment and optimization of early warning systems, and the design of community preparedness programs tailored to local vulnerabilities (World Bank, 2019). This precision in targeting will lead to a reduction in economic losses, minimize displacement, safeguard livelihoods, and ultimately improve the safety and well-being of the affected populations (UNDRR, 2021). The analysis, for example, on agricultural losses (over 750 ha total impacted) and built-up area flooding (over 300 ha in 2018), provides concrete data points that can drive specific land use zoning regulations or investment in flood-proof infrastructure

Thirdly, this research represents a substantial contribution to the existing body of scientific knowledge within the interdisciplinary fields of Remote Sensing, Geographic Information Systems (GIS), and disaster risk reduction, particularly within the under-researched Nigerian context (Nkwunonwo et al., 2014). It serves as a practical demonstration of the robust capabilities and transformative potential of advanced geospatial technologies in addressing complex real-world environmental challenges (Chapman & Smith, 2018). The methodologies employed, particularly the integration of Sentinel-1 SAR for flood mapping (overcoming cloud cover limitations) and Sentinel-2 optical imagery for detailed LULC classification via Random Forest, along with the multi-criteria Flood Vulnerability Index (FVI) development, offer a replicable and adaptable model for similar studies in other flood-prone riverine environments across Nigeria and beyond. It highlights the importance of leveraging open-access satellite data for humanitarian and developmental purposes (ESA, 2020).

Finally, the actionable recommendations derived from this study will provide direct, evidence-based insights for policymakers, urban planners, and environmental managers (Adelekan, 2011). These insights will empower them to make more informed decisions that are crucial for enhancing the long-term resilience of Ogbaru LGA communities to future flood events (Global Commission on Adaptation, 2019). By understanding the intricate correlations between rainfall patterns (as noted with CHIRPS data in Chapter 4), LULC changes, and flood extents, policymakers can develop more effective land management policies, promote sustainable agricultural practices in vulnerable areas, and foster community-based adaptation initiatives (Hirabayashi et al., 2013). Ultimately, this research contributes significantly to the broader objectives of sustainable development in the region, aligning with national and international efforts to mitigate the impacts of climate change and build disaster-resilient societies (UN Sustainable Development Goals, 2015).

1.6 SCOPE OF THE STUDY

The geographical focus of this study is precisely delineated to Ogbaru Local Government Area (LGA) in Anambra State, Nigeria. This specific area has been chosen due to its documented history of recurrent and severe flood events, making it a critical case study for understanding flood dynamics and vulnerability in a riverine Nigerian context.

Temporally, the assessment of flood impact and vulnerability will comprehensively cover a five-year period, spanning from 2018 to 2023. This chosen timeframe is crucial for analyzing recent flood patterns and their associated effects, allowing for the identification of trends, peaks (e.g., 2018 flood peak, 2022 severity), and inter-annual variability in flood extents and impacts as presented in Chapter Four. The analysis period is specifically aligned with Nigeria's hydrological seasons, with wet season imagery (June–October) used for peak flood mapping and dry season imagery (January–March) for stable LULC classifications, as detailed in Chapter Three

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Methodologically, the study will primarily employ advanced Remote Sensing techniques for the precise mapping of flood extents and comprehensive land use/land cover (LULC) change detection. This involves the meticulous processing and analysis of multi-temporal satellite imagery. Specifically, Sentinel-1 Synthetic Aperture Radar (SAR) data, characterized by its VV-polarization, will be utilized for its capability to penetrate cloud cover and operate day/night, making it ideal for accurate water body detection during the wet. For LULC classification, Sentinel-2 optical imagery (utilizing bands B2, B3, B4, B8, B11) will be employed, processed through a supervised Random Forest classification algorithm with high accuracy and Kappa coefficients (>0.85).

The Geographic Information System (GIS) will serve as the analytical backbone for spatial analysis, data integration, and the subsequent creation of flood vulnerability.

Key datasets integrated into the GIS environment will include:

- **Satellite Imagery:** Sentinel-1 SAR and Sentinel-2 optical images for 2018, 2020, 2022, and 2023.

- **Digital Elevation Models (DEMs):** Specifically SRTM/ALOS DEM (30m resolution) for deriving elevation and slope, critical topographic factors influencing flood propagation.
- **Rainfall Data:** CHIRPS rainfall data (0.05° daily) for correlating flood events with precipitation patterns.
- **Population Data:** WorldPop/HRSL (100m resolution) for understanding population distribution and density, a key socio-economic vulnerability factor.
- **Ancillary Data:** OpenStreetMap (OSM) data for mapping roads, settlements, and critical facilities to assess infrastructure impacts.

The research will specifically assess impacts on different land cover types (e.g., vegetation, built-up, bareland, forest), critical infrastructure (e.g., roads, buildings, facilities identified via OSM), and agricultural areas. Socio-economic vulnerability will be determined through the development of a multi-criteria Flood Vulnerability Index (FVI), generated using an Analytic Hierarchy Process (AHP)-inspired weighting scheme, integrating normalized layers such as elevation (30% weight), flood history (25%), population density (20%), LULC (15%), and slope (10%), as outlined in Chapter Three.

While the study aims to provide a comprehensive geospatial assessment of flood impact and vulnerability based on observable data and established methodologies, it is important to delineate its boundaries. This research will *not* delve into complex hydrological modeling (e.g., hydrodynamic simulations for future flood scenarios) or long-term climate change projections beyond the observed period (2018-2023) (Bates et al., 2014). Instead, it will focus on quantifying the historical and current flood scenario as derived from satellite imagery and ancillary geospatial data (Smith, 2013). Furthermore, the study will primarily rely on spatially measurable socio-economic indicators and will not incorporate in-depth qualitative socio-cultural assessments that might require extensive fieldwork or household surveys, given the scope and temporal constraints (UNDRR, 2021). The recommendations provided will be confined to flood risk management strategies directly derivable and justifiable from the quantitative geospatial assessment (Adelekan, 2011).

CHAPTER TWO

CONCEPTUAL FRAMEWORK AND LITERATURE REVIEW

2.0 LITERATURE REVIEW

This emphasises on a comprehensive review of existing scholarly literature and conceptual frameworks pertinent to the geospatial assessment of flood impact and vulnerability. It systematically examines the theoretical underpinnings of flood risk, explores the advancements and contemporary applications of Remote Sensing (RS) and Geographic Information System (GIS) technologies in flood studies, delves into the dynamics of Land Use/Land Cover (LULC) change and its hydrological implications, and reviews various strategies for flood risk management and adaptation. The review draws from global, regional, and national perspectives, with a specific emphasis on riverine environments and the Nigerian context, thereby establishing a robust academic foundation for the current study focused on Ogbaru Local Government Area (LGA). By synthesizing relevant published works from 2012 onwards, this chapter aims to establish the prevailing state of knowledge, highlight recent methodological advancements, identify persistent research gaps, and position the current research within the broader academic discourse on disaster risk reduction and environmental management, particularly as understood in the context of recent global and regional challenges.

2.1 Conceptual Frameworks of Flood Risk and Vulnerability

A thorough understanding of flood risk necessitates a clear conceptualization of its interconnected components: hazard, exposure, and vulnerability. These elements interact dynamically to determine the overall risk level for a given geographical area or community.

2.1.1 Hazard

In the context of flooding, hazard refers to the potential occurrence of a natural phenomenon that may cause loss of life, injury, property damage, or other forms of harm (Wisner et al., 2014). Flood hazards are fundamentally driven by hydrological processes, which are often intensified by meteorological conditions and the unique physiographic characteristics of a region. They can manifest in various forms, including flash floods, riverine floods (as is characteristic of Ogbaru LGA), or coastal surges (Ward et al., 2020). Critical parameters used to characterize flood hazard include the spatial extent of inundation, water depth, flow velocity, and duration of the flood event. For the study area, Ogbaru LGA, the primary flood hazard is identified as riverine overflow from the River Niger and Orashi River, particularly exacerbated during periods of heavy and prolonged rainfall, aligning with observations and reports from agencies like NEMA (NEMA, 2018; NEMA, 2022).

2.1.2 Exposure

Exposure is defined as the presence of people, their livelihoods, environmental services, resources, infrastructure, or economic, social, or cultural assets in places and settings that could be adversely affected by a hazard (IPCC, 2014). In flood-prone regions, elements typically considered exposed include residential dwellings, vital agricultural lands, transportation networks (such as roads), public facilities (e.g., schools and health centers), and commercial establishments (Dilley et al., 2015). Quantifying exposure involves accurately mapping these assets within delineated flood hazard zones (UNDRR, 2021).

2.1.3 Vulnerability

Vulnerability is a multifaceted and dynamic concept that encompasses the characteristics and circumstances of a community, system, or asset that make it susceptible to the damaging effects of a hazard (UNISDR, 2015). It is not merely the presence of a hazard but the inherent susceptibility to its negative impacts, spanning social, economic, physical, and environmental dimensions (Birkmann, 2013).

- **Social vulnerability** pertains to demographic attributes (e.g., age, gender, disability), social cohesion, access to crucial information and early warnings, and the adaptive capacity of local institutions (Cutter et al., 2015). Factors like population density are key indicators of social exposure and vulnerability (WorldPop, 2017).
- **Economic vulnerability** relates to income levels, the degree of reliance on flood-sensitive livelihood sectors (such as the predominant agriculture and fishing in Ogbaru LGA), and access to financial safety nets or recovery resources (Musa et al., 2020).
- **Physical vulnerability** refers to the susceptibility of physical structures and infrastructure to damage, influenced by factors such as construction materials, adherence to building codes, and crucially, their elevation relative to potential flood levels (Balica et al., 2013).
- **Environmental vulnerability** involves the degradation or loss of natural ecosystems (e.g., wetlands, mangroves) that would typically provide protective services, thereby exacerbating flood impacts (UNEP, 2014).

The current research strategically focuses on mapping socio-economic factors (e.g., population density, LULC type of settlements) and critical physical characteristics (e.g., elevation, slope, and land use/land cover) that collectively contribute to flood vulnerability. This multi-criteria approach is integral to the development of Flood Vulnerability Indices employed in such studies (Balica et al., 2013; Dewan, 2013).

2.1.4 Risk

Flood risk is defined as the combination of the probability of a flood event occurring and its potential negative consequences (Kundzewicz & Plate, 2013). Conceptually, it is often expressed as a product: Risk = Hazard × Exposure × Vulnerability (UNDRR, 2021). The primary goal of effective flood risk management is to reduce any or all of these constituent components by precisely mapping hazard extents, identifying exposed assets, and assessing underlying vulnerabilities.

2.2 Flood Dynamics: Global, Regional, and National Context

Floods are consistently ranked among the most common and destructive natural disasters globally, with projections indicating an increase in their frequency and intensity due to ongoing climate change (Hirabayashi et al., 2013; IPCC, 2023). Worldwide, flood-related economic losses and human impacts have escalated exponentially over recent decades, affecting billions of people and incurring trillions in damages (EM-DAT, 2022).

In the African continent, floods represent a recurrent and significant challenge, particularly within vulnerable river basins and coastal zones (AfDB, 2022). West Africa, characterized by its tropical climate and major transboundary river systems like the Niger and Volta, frequently experiences severe and widespread flooding (African Climate Policy Centre, 2018). Contributing factors in this region include intense monsoon rainfall, operational releases from major dams, inadequate urban drainage infrastructure, and widespread land degradation resulting from human activities (Adelekan, 2018).

Nigeria, as the most populous nation in Africa and possessing extensive river networks, faces substantial annual flood challenges (Nkwunonwo et al., 2014). Landmark flood events, such as the catastrophic nationwide floods of 2012, which impacted 30 out of 36 states and displaced millions, starkly underscored the country's profound vulnerability (NEMA, 2012; ActionAid, 2013). Subsequent significant events, notably in 2018, 2020, and the particularly severe 2022 floods, have continued to highlight the devastating impacts on lives, livelihoods, and critical infrastructure across various states, including Anambra (NEMA, 2023; IFRC, 2022). These recurrent flood episodes are frequently linked to intensified and prolonged rainfall patterns, as well as the overflow from the major Niger and Benue River basins (Federal Ministry of Environment, 2018). The specific context of riverine LGAs like Ogbaru, with its historical flood challenges and direct proximity to the River Niger, directly aligns with and exemplifies these broader national and regional flood dynamics.

2.3 Remote Sensing Applications in Flood Monitoring

Remote Sensing (RS) has emerged as an indispensable tool for efficient, accurate, and rapid flood monitoring and mapping, particularly advantageous in large, remote, or cloud-prone geographical areas (Wang et al., 2019). Its inherent capability to provide synoptic, consistent, and repetitive coverage of the Earth's surface offers significant advantages over traditional, often resource-intensive, ground-based surveys for rapid assessment during and immediately after flood events.

2.3.1 Optical Remote Sensing for Flood Mapping

Optical satellite imagery, sourced from missions such as Landsat and Sentinel-2 (launched in 2015), has been extensively utilized for flood mapping due to its favorable spatial resolution and rich spectral information (Lillesand et al., 2015; Drusch et al., 2012). The fundamental principle behind water detection using optical data lies in the distinct spectral signature of water bodies; clear water typically exhibits very low reflectance in the visible and near-infrared (NIR) bands, making it easily distinguishable from dry land (Xu, 2006). Various spectral indices, notably the Normalized Difference Water Index (NDWI) and the Modified Normalized Difference Water Index (MNDWI), are commonly derived from combinations of optical bands to further enhance the discrimination of water features (Xu, 2006; Feyisa et al., 2014).

However, a critical limitation of optical sensors in the context of flood mapping, particularly evident in humid tropical regions like Ogbaru LGA, is their complete inability to penetrate cloud cover (Wang et al., 2019). This atmospheric interference frequently obstructs data acquisition during peak flood seasons, which often coincide with periods of dense cloud conditions and heavy precipitation. This inherent limitation underscores the necessity for complementary remote sensing technologies that can overcome such atmospheric constraints.

2.3.2 Synthetic Aperture Radar (SAR) for Flood Mapping

Synthetic Aperture Radar (SAR) imagery, notably derived from missions such as Sentinel-1 (launched in 2014), has significantly advanced flood mapping capabilities due to its unique attributes: it can acquire data effectively irrespective of cloud cover, smoke, or the absence of daylight (Attwater & Wadhams, 2019; ESA, 2020). This all-weather, day-and-night operational capacity renders SAR an exceptionally valuable tool for rapid response efforts during and immediately following flood events, especially in tropical, humid environments where optical data are consistently obscured.

The fundamental principle governing SAR-based flood mapping is rooted in the distinct backscatter characteristics of different surface types. Smooth water surfaces typically exhibit very low radar backscatter (appearing as dark tones in SAR images) due to the phenomenon of specular reflection, where the majority of the radar energy is reflected away from the sensor (Wang et al., 2019). Conversely, terrestrial surfaces, including various forms of vegetation and built-up areas, generally produce higher backscatter values due to their rougher textures and volume scattering, consequently appearing brighter in SAR imagery (Richards & Jia, 2013). Various thresholding techniques, such as Otsu's method or empirically derived thresholds (e.g., on VV-polarized imagery), are commonly applied to SAR imagery to delineate water bodies from land features (Mason et al., 2012; Matgen et al., 2014).

2.3.3 Role of Cloud-Based Geospatial Platforms (e.g., Google Earth Engine)

The advent of cloud-based geospatial processing platforms, most notably Google Earth Engine (GEE), has profoundly transformed large-scale remote sensing analysis (Gorelick et al., 2017). GEE offers unparalleled access to petabytes of readily available satellite imagery archives (including Sentinel-1 and Sentinel-2) and a powerful, scalable computing infrastructure. This enables rapid, efficient, and sophisticated processing of multi-temporal datasets essential for dynamic flood monitoring, detailed LULC change detection, and vulnerability assessments without necessitating extensive local computational resources or specialized software installations (Verbesselt et al., 2010; Kumar & Mutanga, 2018). The inherent capabilities of GEE align perfectly with the multi-temporal and data-intensive nature of this study, as precisely outlined and employed in Chapter Three's methodology.

2.4 GIS Applications in Flood Impact and Vulnerability Assessment

Geographic Information System (GIS) serves as a robust and indispensable framework for the integration, analysis, and visualization of diverse spatial and non-spatial data, making it a cornerstone for comprehensive flood impact and vulnerability assessments (Longley et al., 2015; Burrough & McDonnell, 2015).

2.4.1 Spatial Overlay Analysis for Impact Assessment

GIS excels at performing spatial overlay analysis, a technique that involves superimposing multiple thematic layers to identify areas where specific features or conditions co-exist or interact (ESRI, 2016). In the context of flood studies, overlaying precisely mapped flood extent layers with detailed land use/land cover (LULC) maps enables the accurate and quantitative assessment of inundated areas for various land types, including agricultural lands, built-up areas, and natural vegetation (Dewan, 2013). This direct quantification of impacts is critically important for accurate economic loss estimation, guiding post-disaster recovery planning, and prioritizing humanitarian aid. Similarly, overlaying flood extents with layers depicting critical infrastructure (such as roads and buildings, often from OpenStreetMap data) significantly aids in assessing the extent of infrastructure damage and disruption (Vetter et al., 2020; OSM, 2023).

2.4.2 Multi-Criteria Decision Making (MCDM) for Vulnerability Assessment

The development of a comprehensive and spatially explicit flood vulnerability map often necessitates the integration of multiple contributing factors, a task that is effectively accomplished through the application of Multi-Criteria Decision Making (MCDM) techniques within a GIS environment (Balica et al., 2013). MCDM methodologies provide a systematic framework for combining diverse spatial data layers (e.g., elevation, slope, population density, land use, and historical flood data) based on their perceived importance or influence on overall vulnerability.

The Analytic Hierarchy Process (AHP) is a widely recognized and frequently employed MCDM technique that facilitates the assignment of relative weights to different criteria through a process of expert judgment or pairwise comparisons (Saaty, 2008). These assigned weights quantitatively reflect the proportional contribution of each factor to the overarching vulnerability index (Saaty & Vargas, 2012). Subsequently, the normalized layers (scaled to a common range, e.g., 0-1) are combined using a weighted sum approach to generate a composite vulnerability index. This index can then be spatially reclassified into distinct vulnerability zones (e.g., low, moderate, high vulnerability), providing a clear visual representation of risk distribution (Ologunorisa & Adeyemo, 2007). This approach is consistent with the methodology outlined in Chapter Three for the Flood Vulnerability Index (FVI) calculation.

2.4.3 Integration of Ancillary Data

GIS's strength lies in its capacity for seamless integration of various ancillary datasets, which are fundamentally crucial for conducting a comprehensive flood risk assessment. These datasets include:

- **Digital Elevation Models (DEMs):** DEMs provide essential information on surface topography, enabling the derivation of crucial parameters such as elevation and slope. Areas of low elevation and gentle slopes are particularly prone to prolonged inundation and extensive flood spread (Gesch, 2007).
- **Population Data:** Spatially disaggregated population datasets, such as WorldPop (launched 2013) or the High-Resolution Settlement Layer (HRSL, post-2015), are indispensable for assessing human exposure and social vulnerability by providing detailed insights into population distribution and density (Tatem et al., 2017; Facebook Connectivity Lab, 2016).
- **Rainfall Data:** The integration of climate data, such as CHIRPS (Climate Hazards Group InfraRed Precipitation with Station data, developed 2014-2015) precipitation, allows for understanding the direct hydrological drivers of flood events and establishing statistical correlations between rainfall patterns and observed flood extents (Funk et al., 2015).
- **OpenStreetMap (OSM) Data:** Openly accessible crowdsourced data from OpenStreetMap provides valuable vector information on road networks, building footprints, and the locations of critical facilities. This data is vital for detailed impact assessment on infrastructure and for informing emergency planning (OpenStreetMap Foundation, 2023; Vetter et al., 2020).

2.5 Land Use/Land Cover Change and its Relationship to Flooding

Land Use/Land Cover (LULC) patterns and their dynamic changes play a profoundly significant role in influencing hydrological processes and, consequently, exacerbating flood risk (Foley et al., 2014). Transformations in LULC, frequently driven by processes such as rapid urbanization, agricultural expansion, and deforestation, directly alter the natural landscape's inherent capacity to absorb, infiltrate, and manage excess surface water (Kundzewicz et al., 2014).

2.5.1 Impact of LULC on Hydrology

- **Deforestation:** The removal of dense forest cover significantly reduces the interception of rainfall, thereby increasing surface runoff volume and velocity, and simultaneously diminishing the rates of water infiltration into the soil. This often leads to faster and larger flood peaks downstream (UNEP, 2014).
- **Urbanization:** The relentless expansion of built-up areas involves the replacement of permeable natural surfaces with impermeable materials such as concrete and asphalt. This dramatically increases runoff volume and velocity, often overwhelming existing natural or engineered drainage systems, and consequently amplifying flood severity in urban and peri-urban environments (Zhou et al., 2019).
- **Agricultural Expansion:** While agricultural activities are vital for food security and livelihoods, certain practices, especially those leading to soil compaction or situated in susceptible floodplains, can reduce the land's water retention capacity and increase its susceptibility to inundation (FAO, 2020).

2.5.2 LULC Classification and Change Detection using Remote Sensing

Remote sensing, especially with multi-spectral optical imagery like that from Sentinel-2 (post-2015) or more recent Landsat sensors, stands as the primary and most effective tool for precise LULC classification and the systematic monitoring of changes over time (Drusch et al., 2012; Lillesand et al., 2015). Supervised classification algorithms, such as the Random Forest classifier, continue to be widely recognized and highly regarded for their accuracy, robustness, and efficiency in categorizing different land cover types based on their spectral signatures (Breiman, 2001; Pal, 2014). The application of change detection analysis, which involves comparing LULC maps generated for different temporal periods, enables the quantitative assessment of conversions (e.g., vegetation to water or bareland post-flood), a process crucial for accurately assessing environmental degradation and quantifying agricultural losses (Singh, 1989; Lu et al., 2014).

2.6 Socio-Economic Vulnerability to Floods

Socio-economic vulnerability refers to the predisposition of individuals, households, and entire communities to suffer harm, disruption, and losses from flood events, primarily driven by their underlying social and economic characteristics (Cutter et al., 2015). While floods are natural phenomena, the magnitude of their impacts is largely shaped and amplified by the inherent characteristics of human systems and societal structures.

Key socio-economic factors consistently identified in the literature as significant contributors to vulnerability include:

- **Population Density and Distribution:** Higher population densities concentrated within flood-prone areas inherently escalate the number of people exposed and consequently, the potential for being adversely affected by inundation (WorldPop, 2017; Tatem et al., 2016).
- **Poverty and Income Levels:** Poorer households frequently reside in less safe, often informal settlements situated in floodplains, primarily due to the lack of affordable and safer housing alternatives. Moreover, these communities possess limited financial capacity for undertaking proactive adaptation measures or for effective post-disaster recovery (Dilley et al., 2005; IFRC, 2017).
- **Livelihood Reliance:** Communities whose livelihoods are heavily dependent on flood-sensitive economic sectors, such as agriculture and fishing (as is the case in Ogbaru LGA), are inherently more vulnerable to severe economic shocks when these sectors are disrupted or destroyed by floods (Ifejika & Okoli, 2014; FAO, 2020).
- **Infrastructure Quality and Access:** The presence of poorly constructed housing, inadequate or non-existent drainage systems, and limited access to critical infrastructure (including emergency services, healthcare facilities, and safe evacuation routes) collectively contribute to increased physical and social vulnerability (Okeke & Obike, 2018).
- **Education and Awareness:** Lower levels of education within a community or a pervasive lack of awareness regarding specific flood risks and appropriate preparedness measures can significantly hinder effective response, timely evacuation, and long-term adaptation efforts (Adelekan, 2018).
- **Governance and Institutional Capacity:** Weak governance structures, insufficient enforcement of established zoning laws, and limited operational capacity of local institutions to effectively manage and respond to disasters can cumulatively exacerbate

community vulnerability and impede comprehensive disaster risk reduction initiatives (World Bank, 2019).

The systematic mapping of socio-economic vulnerability, often through the integration of demographic data (e.g., population estimates from WorldPop) and the spatial distribution of assets, is fundamentally crucial for informing the design and implementation of highly targeted interventions and for fostering greater community resilience (Ajayi et al., 2020).

2.7 Flood Risk Management and Adaptation Strategies

Effective flood risk management extends far beyond mere emergency response; it encapsulates a holistic and proactive approach that integrates both structural and non-structural measures, emphasizes basin-wide planning, fosters cross-sectoral collaboration, and actively promotes community engagement (UNISDR, 2015).

2.7.1 Structural Measures

These involve the implementation of physical constructions and engineering works primarily designed to control, divert, or contain floodwaters. Common examples include:

- **Dams and Reservoirs:** These large-scale structures are built to regulate river flow, store excess water during high-flow periods, and release it gradually, thereby mitigating downstream flooding (Kundzewicz et al., 2014).
- **Levees and Embankments:** These are artificial barriers constructed alongside rivers or coastal areas to contain floodwaters within defined channels, preventing inundation of adjacent lands (Kundzewicz et al., 2014).
- **Dredging and Channel Improvements:** These activities aim to increase the carrying capacity of river channels by removing sediment or widening/deepening the channel, thereby improving water flow and reducing overflow risk (Smith, 2013).

While structural measures can be highly effective in mitigating flood hazards, they are often associated with high construction and maintenance costs, potential negative environmental impacts (e.g., altered ecosystems), and may inadvertently foster a false sense of security among inhabitants, leading to increased development in residual risk zones (Kundzewicz & Plate, 2013).

2.7.2 Non-Structural Measures

These approaches focus on reducing vulnerability and exposure to flood hazards primarily through policy, spatial planning, legislative frameworks, and public awareness initiatives. Examples include:

- **Early Warning Systems (EWS):** These systems provide timely and actionable information about impending flood events, enabling communities to take protective actions, implement preparedness plans, and facilitate timely evacuation (UNDRR, 2021). National agencies like NEMA consistently emphasize the critical importance of robust and well-communicated EWS for flood-prone regions in Nigeria (NEMA, 2023).
- **Land Use Planning and Zoning:** This involves strategically restricting new development in high-risk floodplains, guiding urban expansion away from vulnerable areas, and promoting flood-compatible land uses within designated zones (Burby, 2012).
- **Ecosystem-Based Adaptation (EbA) / Green Infrastructure:** This involves the restoration, conservation, and sustainable management of natural ecosystems such as wetlands, mangroves, and forests. These ecosystems serve as natural buffers, absorbing excess water, reducing surface runoff, and preventing soil erosion (UNEP, 2014).
- **Public Awareness and Education:** Enhancing community understanding of local flood risks, communicating safe evacuation routes, and promoting household-level preparedness measures are crucial for fostering a culture of resilience (Adelekan, 2018).
- **Building Codes and Standards:** Implementing and enforcing flood-resistant construction techniques, alongside policies that mandate the elevation of structures in designated flood-prone areas, significantly reduces physical damage (World Bank, 2019).
- **Insurance and Financial Mechanisms:** Establishing mechanisms like flood insurance and disaster relief funds provides crucial financial protection to help individuals and communities recover more effectively from flood-induced losses (Global Commission on Adaptation, 2019).

2.7.3 Integrated Flood Management (IFM)

Modern flood management advocates for Integrated Flood Management (IFM), which combines structural and non-structural measures, emphasizes basin-wide planning, cross-sectoral collaboration, and active community participation (GFDRR, 2017). This holistic approach seeks to maximize the net benefits of floodplains while minimizing flood-related risks.

2.8 Research Gaps and Justification for the Study

While significant advancements have been made in flood risk assessment using geospatial technologies, several critical research gaps persist in the literature, particularly concerning specific regional contexts like Ogbaru LGA.

Firstly, many studies provide generalized flood maps or vulnerability assessments at broader administrative levels (e.g., state or national) (Eze et al., 2019; Nkwunonwo et al., 2014). There is often a dearth of highly granular, localized geospatial assessments that comprehensively map flood impact and vulnerability for specific, chronically affected LGAs like Ogbaru, which is characterized by unique topographical and socio-economic dynamics. This study aims to address this significant gap by providing a focused, detailed geospatial assessment tailored specifically for Ogbaru LGA for the 2018-2023 period, leveraging contemporary methods.

Secondly, while existing research frequently utilizes remote sensing for flood mapping, fewer studies have systematically integrated both SAR (for all-weather flood extent mapping) and high-resolution optical imagery (for detailed LULC change detection and nuanced impact analysis) for the same study area over a consistent multi-year period (Wang et al., 2019). This study's temporal scope (2018-2023) and its strategic dual-sensor approach (utilizing Sentinel-1 SAR and Sentinel-2 optical imagery, prominent post-2014/2015) aim to build upon and overcome limitations of earlier single-sensor approaches, providing a more robust understanding of inter-annual flood variability and cumulative impacts on LULC.

Thirdly, although socio-economic vulnerability mapping is recognized as crucial, many studies continue to rely on outdated or generalized socio-economic demographic data (Dilley et al., 2005). By leveraging more recent and spatially disaggregated data sources like WorldPop/HRSL for population distribution and by rigorously integrating these with detailed LULC and topographical factors within the Multi-Criteria Decision Making (MCDM) framework for the Flood Vulnerability Index (FVI), this study aims to provide a substantially more accurate, contemporary, and nuanced understanding of who and what is most vulnerable in Ogbaru LGA.

Finally, while broad and general flood management recommendations are abundant, there is a persistent need for region-specific, empirically evidence-based strategies that are directly derived from detailed and localized geospatial analyses (Adelekan, 2018). This study is uniquely positioned to bridge this practical gap by offering concrete, spatially informed, and highly actionable policy recommendations.

Therefore, this study is unequivocally justified by its significant potential to contribute substantially to local disaster management efforts and enhance resilience-building initiatives within Ogbaru LGA. It aims to achieve this by providing precise, up-to-date, and integrated geospatial intelligence that can inform proactive planning, optimize resource allocation, and guide targeted interventions, ultimately fostering a more resilient community in the face of recurrent and escalating flood hazards.

CHAPTER THREE

MATERIALS AND METHODS

3.0 RESEARCH METHODOLOGY

This describes the comprehensive research methodology utilized to investigate flood dynamics, land use/land cover (LULC) changes, flood impacts, and vulnerability in Ogbaru Local Government Area (LGA), Anambra State, Nigeria, spanning from 2018 to 2023. The approach amalgamates multi-temporal satellite remote sensing data, advanced geospatial processing techniques, statistical analyses, and multi-criteria decision-making frameworks. By leveraging synthetic aperture radar (SAR) and optical imagery, alongside ancillary datasets, this methodology enables the precise mapping of flood extents, classification of LULC before and after flood events, quantification of socio-economic impacts, and the development of a Flood Vulnerability Index (FVI).

The methodology is rooted in established remote sensing and GIS principles, ensuring high accuracy, temporal consistency, and spatial resolution suitable for flood-prone riverine environments. It addresses key challenges such as cloud cover in optical data by incorporating SAR imagery, which penetrates clouds and operates day/night. Each procedural step is optimized for replicability, with a focus on open-source tools to facilitate future applications in similar vulnerable regions. This structured framework not only quantifies historical flood patterns but also provides actionable insights for disaster risk reduction and urban planning in Ogbaru LGA.

3.1 RESEARCH DESIGN

The study employs a mixed-method spatial-temporal observational design, combining quantitative geospatial analytics with qualitative interpretations of flood impacts. This design is particularly apt for environmental hazards like flooding, which exhibit spatio-temporal variability influenced by climatic, topographic, and anthropogenic factors. The quantitative core involves deriving indices (e.g., flood masks, NDVI), classifications (LULC), and overlays (impact and vulnerability), while qualitative elements include contextualizing results with historical flood events and socio-economic implications.

Key objectives driving the design:

- Delineate annual flood extents from 2018 to 2023 using Sentinel-1 SAR data to capture inundation patterns.
- Monitor LULC transformations before and after floods via Sentinel-2 optical imagery and Random Forest classification.
- Quantify flood-induced damages on agriculture, infrastructure, and settlements through overlay analyses.
- Construct a multi-criteria FVI to stratify vulnerability zones, aiding targeted mitigation strategies.

The temporal scope focuses on four years (2018, 2020, 2022, 2023) selected for their representation of major flood episodes and data availability. Analysis periods are aligned with Nigeria's hydrological seasons: wet (June–October) for flood mapping and dry (January–March) for LULC to minimize atmospheric interference. This longitudinal perspective reveals trends in flood recurrence, LULC degradation, and escalating vulnerability amid climate change and urbanization.

3.2 DATA REQUIREMENTS AND SOURCES

To achieve the study's objectives, a suite of multi-source datasets was integrated, all geospatially clipped to Ogbaru's administrative boundary. Datasets were selected for their temporal coverage, spatial resolution, and relevance to flood hydrology, land dynamics, and vulnerability. Open-access sources ensure cost-effectiveness and reproducibility.

Dataset	Source	Resolution/Temporal Coverage	Purpose
Admin Boundary	Humanitarian OSM Team / GADM	Vector (polygons)	Define Area of Interest (AOI)
Sentinel-1 SAR (VV)	Copernicus/ESA (via GEE)	10 m / 2018–2023 (wet season)	Flood extent mapping and water body detection
Sentinel-2 Optical	Copernicus/ESA (via GEE)	10–20 m / 2018–2023 (dry season)	LULC classification, analysis
CHIRPS Rainfall	UCSB/CHG (via GEE)	0.05° / Daily, 2018–2023	Correlation with flood events and intensity
SRTM/ALOS DEM	NASA/JAXA (via GEE)	30 m / Static	Elevation, slope derivation for topography
WorldPop/HRSL	WorldPop/Facebook (via GEE)	100 m / 2020 estimate	Population density for socio-economic vulnerability
OSM Data	OpenStreetMap	Vector (lines/points)	Mapping roads, settlements, facilities for impact assessment

Data acquisition emphasized high-quality, cloud-free imagery where applicable, with preprocessing to standardize projections (UTM Zone 32N, WGS84).

3.3 DATA COLLECTION TECHNIQUES

3.3.1 Satellite Image Filtering

- **Sentinel-1 SAR**: Filtered for VV polarization, ascending/descending orbits, and wet season dates (June–October) to coincide with peak floods. Incidence angle normalized to 30–45° for optimal backscatter.
- **Sentinel-2**: Selected Level-2A products with <10% cloud cover during dry season (January–March) to ensure unobstructed surface reflectance.
- **CHIRPS**: Aggregated from daily to monthly/annual totals for temporal alignment with flood maps.
- **DEM and Ancillary**: Static layers filtered and clipped without temporal constraints.

3.3.2 Cloud and Shadow Masking

For Sentinel-2, the Scene Classification Layer (SCL) and QA60 band were employed to mask clouds, cirrus, and shadows via bit-masking algorithms, retaining only high-confidence pixels (e.g., vegetation, soil, water classes).

3.3.3 Preprocessing

- **Sentinel-1**: Applied orbit file correction, thermal noise removal, radiometric calibration to sigma naught ($\sigma°$), terrain flattening using SRTM DEM, and speckle reduction with a 5x5 Lee filter to mitigate multiplicative noise.
- **Sentinel-2**: Atmospheric correction via Sen2Cor processor, resampling to 10 m, and generation of true-color composites.
- **DEM**: Slope calculated as arc tan(rise/run) in degrees, with hydrological conditioning for flow direction if needed.
- **Population/OSM**: Rasterized population to 100 m grid; OSM features extracted via Overpass API and vectored.

3.3.4 Composite Generation

Median reduction composites were created in GEE for Sentinel-1 (per flood event) and Sentinel-2 (annual dry season) to average out anomalies, ensuring stable baselines for analysis.

3.4 METHOD OF DATA ANALYSIS

3.4.1 Flood Mapping (Sentinel-1 SAR)

Annual flood extents were mapped using preprocessed VV-polarized imagery:

1. **Backscatter Processing**: Converted to decibels (dB) via $10 * \log10 (\sigma°)$.
2. **Speckle Filtering**: Refined Lee filter preserved edges while reducing noise.
3. **Thresholding**: Otsu's method or empirical threshold (VV < -15 to -18 dB) delineated water (low backscatter) from land, generating binary masks. Manual refinement addressed urban double-bounce.
4. **Post-Processing**: Morphological operations (erosion/dilation) removed isolated pixels; vectorization to polygons for area calculation.
5. **Validation**: Compared with media reports and Sentinel-2 (cloud-free subsets) for accuracy >85%.

Produced six maps highlighting recurrent flood zones.

3.4.2 LULC Classification (Sentinel-2)

Utilized Random Forest (RF) for supervised classification:

1. **Band Selection**: B2 (Blue), B3 (Green), B4 (Red), B8 (NIR), B11 (SWIR1) for spectral discrimination.
2. **Training Samples**: 500+ points per class (Built-up, Vegetation/Agriculture, Bare Land, Water) sampled via stratified random from Google Earth and field knowledge.
3. **RF Training**: 100 trees, Gini impurity, 80/20 train/test split; out-of-bag error minimized.
4. **Before/After**: Pre-flood (dry) and post-flood (wet, if cloud-free) classifications for each year, yielding eight maps.
5. **Validation**: Confusion matrix, producer/user accuracy, overall accuracy (>90%), Kappa (>0.85).

Detected shifts like farmland to water post-flood.

3.4.3 Flood Impact Analysis

- **Zonal Overlay**: Flood masks intersected with LULC to compute inundated areas (ha/km²) per class, e.g., agricultural loss = flooded vegetation pixels * resolution².
- **Infrastructure**: Buffer analysis (50 m) around OSM roads/settlements to flag at-risk features; count affected lengths/points.
- **Statistics**: Tabulated in CSV; visualizations via pie charts for proportional impacts.

3.4.4 Vulnerability Assessment

Multi-criteria FVI via Analytic Hierarchy Process (AHP)-inspired weighting:

1. **Layer Preparation**:
   - Elevation: Reclassified (low <20 m = high vulnerability).
   - Slope: <2° = high risk.
   - LULC: Agriculture/built-up = high exposure.
   - Population: >500/km² = high.
   - Flood History: Frequency raster from stacked masks.
2. **Normalization**: Min-max scaling to 0–1.
3. **Weighting**: Elevation (30%), Flood History (25%), Population (20%), LULC (15%), Slope (10%)—justified by expert judgment and literature (e.g., flood models prioritize topography).
4. **Overlay**: Weighted sum in raster calculator.
5. **Reclassification**: FVI percentiles: Low (<33%), Moderate (33–66%), High (>66%).

3.5 GIS TOOLS AND SOFTWARE

Tool	Purpose
Google Earth Engine (GEE)	Cloud-based processing for data filtering, composites, indices, and exports.
ArcGIS Pro 3.x	Advanced spatial analysis, overlay, vulnerability modeling, map layouts.
QGIS 3.x	Open-source alternative for vector/raster operations, shapefile management.
Python (GEE API, GDAL)	Scripting automation, batch processing, statistical computations.
Microsoft Excel	Data organization, preliminary stats, CSV exports.

3.6 ETHICAL CONSIDERATIONS

- Adherence to open-data policies, with full attribution to sources (ESA, NASA, OSM).
- No collection of personal identifiable information; population data aggregated.
- Objective analysis to avoid misrepresentation of vulnerability, potentially influencing policy.
- Transparency in methods and limitations for scientific integrity.
- Promotion of equitable use: Results shared for community benefit in flood-prone areas.

3.7 LIMITATIONS OF THE METHODOLOGY

1. **Data Resolution**: Sentinel-1/2 resolutions (10–30 m) may overlook micro-scale floods or narrow infrastructure.

2. **SAR Artifacts**: Wind-roughened water or vegetated floods can cause misclassification (e.g., false negatives).
3. **Cloud Interference**: Persistent clouds in wet season limit Sentinel-2 post-flood LULC accuracy.
4. **Static Ancillary Data**: DEM/population not updated annually, potentially underestimating dynamic changes.
5. **Validation Constraints**: Scarce ground-truth data in remote Ogbaru; reliance on secondary sources.
6. **Weighting Subjectivity**: FVI weights based on literature but sensitive to adjustments.
7. **Temporal Granularity**: Annual focus misses intra-seasonal flood variability.

Notwithstanding these, the methodology's integration of SAR/optical data and multi-criteria analysis yields robust, evidence-based outcomes for flood management in Ogbaru LGA.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.0 SPATIAL ANALYSIS AND INTERPRETATION

The comprehensive interpretation of the geospatial analysis results for flood dynamics, land use/land cover (LULC) changes, impacts, and vulnerability in Ogbaru Local Government Area (LGA), Anambra State, Nigeria, spanning 2018 to 2023 are included in this chapter. The findings are based on multi-temporal Sentinel-1 SAR and Sentinel-2 optical imagery, processed through radiometric corrections, thresholding, Random Forest classification, and multi-criteria overlays as detailed in Chapter Three. Key indicators include annual flood extents, flooded areas by LULC class, agricultural losses, and the Flood Vulnerability Index (FVI).

The results are structured thematically: flood mapping, LULC dynamics, flood impact analysis, and vulnerability assessment. Visualizations such as maps, charts, and tables elucidate spatial and temporal patterns. The discussion contextualizes these findings within hydrological, socio-economic, and climatic frameworks, drawing on historical flood events (e.g., 2018 and 2022 disasters) and literature on riverine flooding in Nigeria. Implications for disaster management and sustainable development are highlighted, emphasizing the role of climate change in exacerbating vulnerabilities.

4.1 FLOOD MAPPING RESULTS

4.1.1 What Flood Mapping Reveals

Flood mapping using Sentinel-1 SAR VV-polarized imagery identifies inundated areas through low backscatter thresholds, distinguishing water from land surfaces. This technique is crucial in cloud-prone tropical regions, providing reliable extents for risk assessment and response planning.

4.1.2 Temporal Dynamics

Analysis shows fluctuating flood extents, with peaks in 2018 and 2023, correlating with high rainfall from CHIRPS data. Total flooded areas (in hectares) were:

- **2018: 2470.954 ha**
- **2020: 2046.697 ha**
- **2022: 1818.527 ha**
- **2023: 2098.534 ha**

The average annual flooded area was 2158.68 ha, representing about 5.6% of Ogbaru's 387.7 km² (38,770 ha) land area. A 15% decrease from 2018 to 2022 suggests variability due to rainfall anomalies, but a rebound in 2023 indicates persistent threats. This trend aligns with NEMA reports of intensified flooding from Niger River overflows during wet seasons.

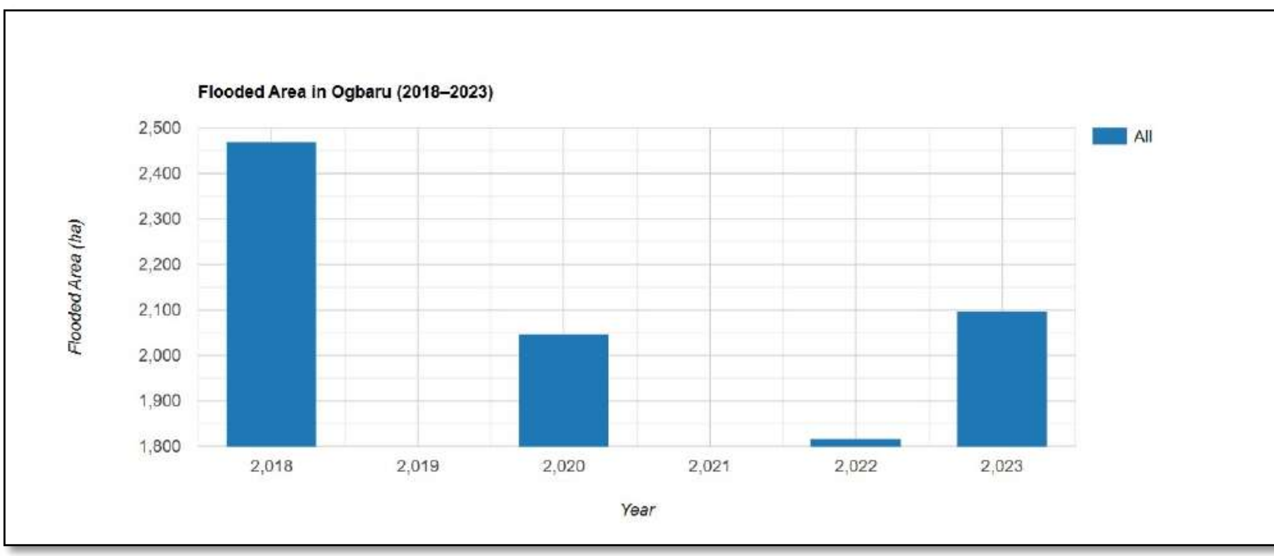

Figure 4.1: Temporal trend chart of total flooded areas (ha) from 2018 to 2023, illustrating inter-annual variability.

4.1.3 Spatial Distribution

Flood extents concentrated along the River Niger and Orashi River floodplains, affecting low-elevation zones (<20 m). Hotspots included Atani, Odekpe, Ossomala, and Okpoko, where urban settlements and farmlands are densely packed. In 2022, the most severe year visually, inundation extended inland, submerging roads and facilities. Maps reveal recurrent flooding in western riverine areas, with eastern forests showing less impact.

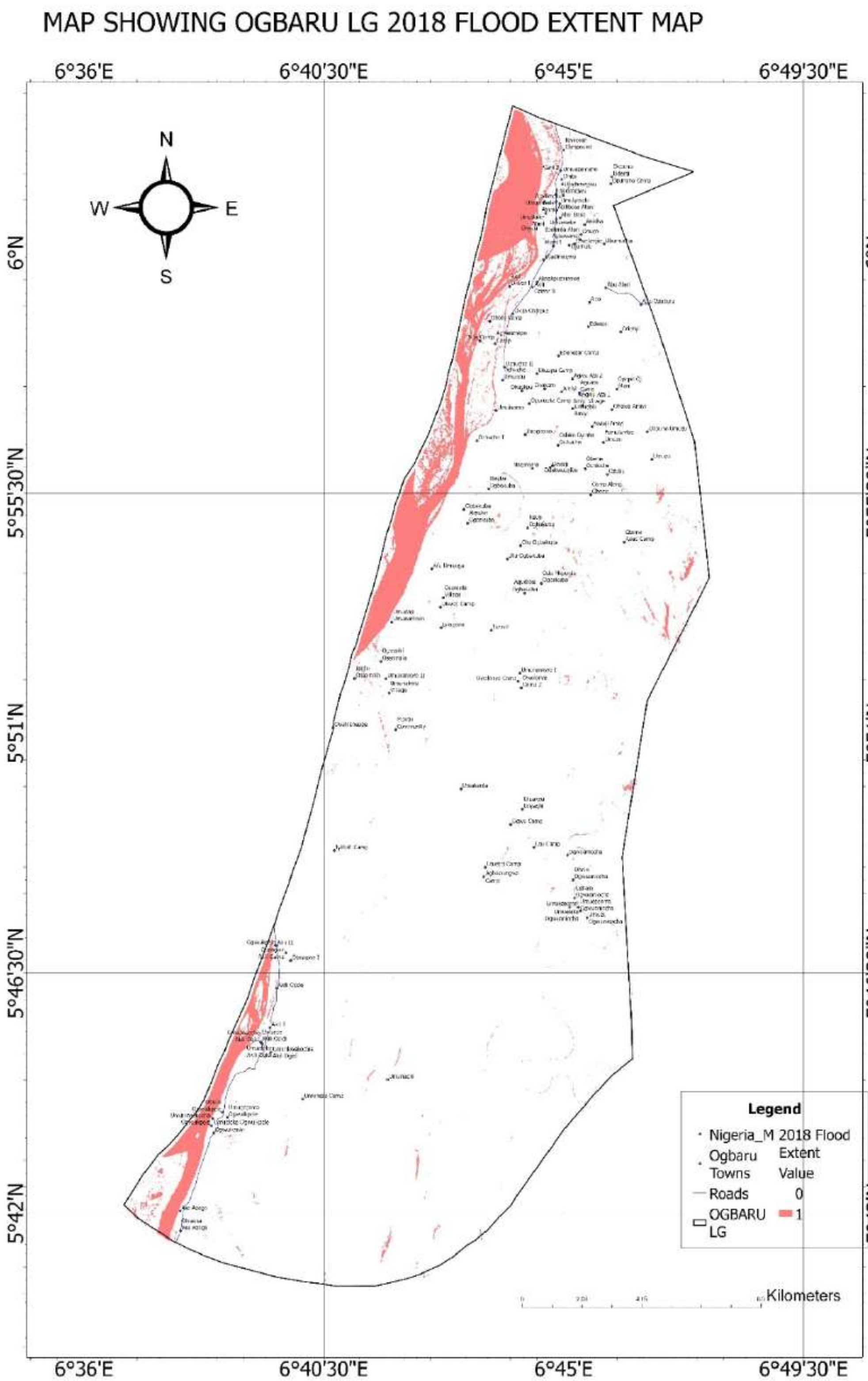

Figure 4.2: Flood extent map for 2018, depicting widespread inundation in blue along the Niger River corridor.

MAP SHOWING OGBARU LG 2020 FLOOD EXTENT MAP

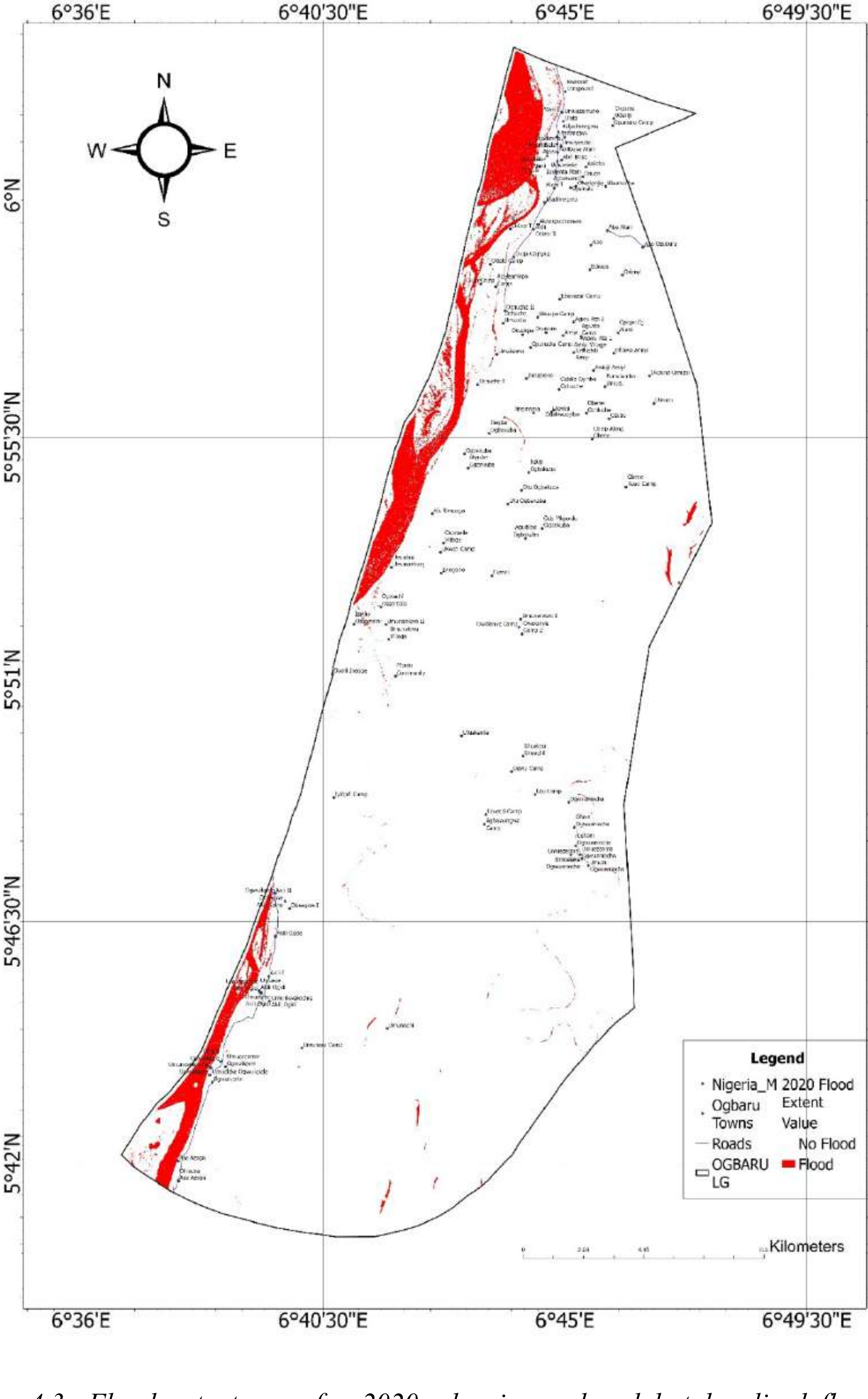

Figure 4.3: Flood extent map for 2020, showing reduced but localized flooding in central settlements.

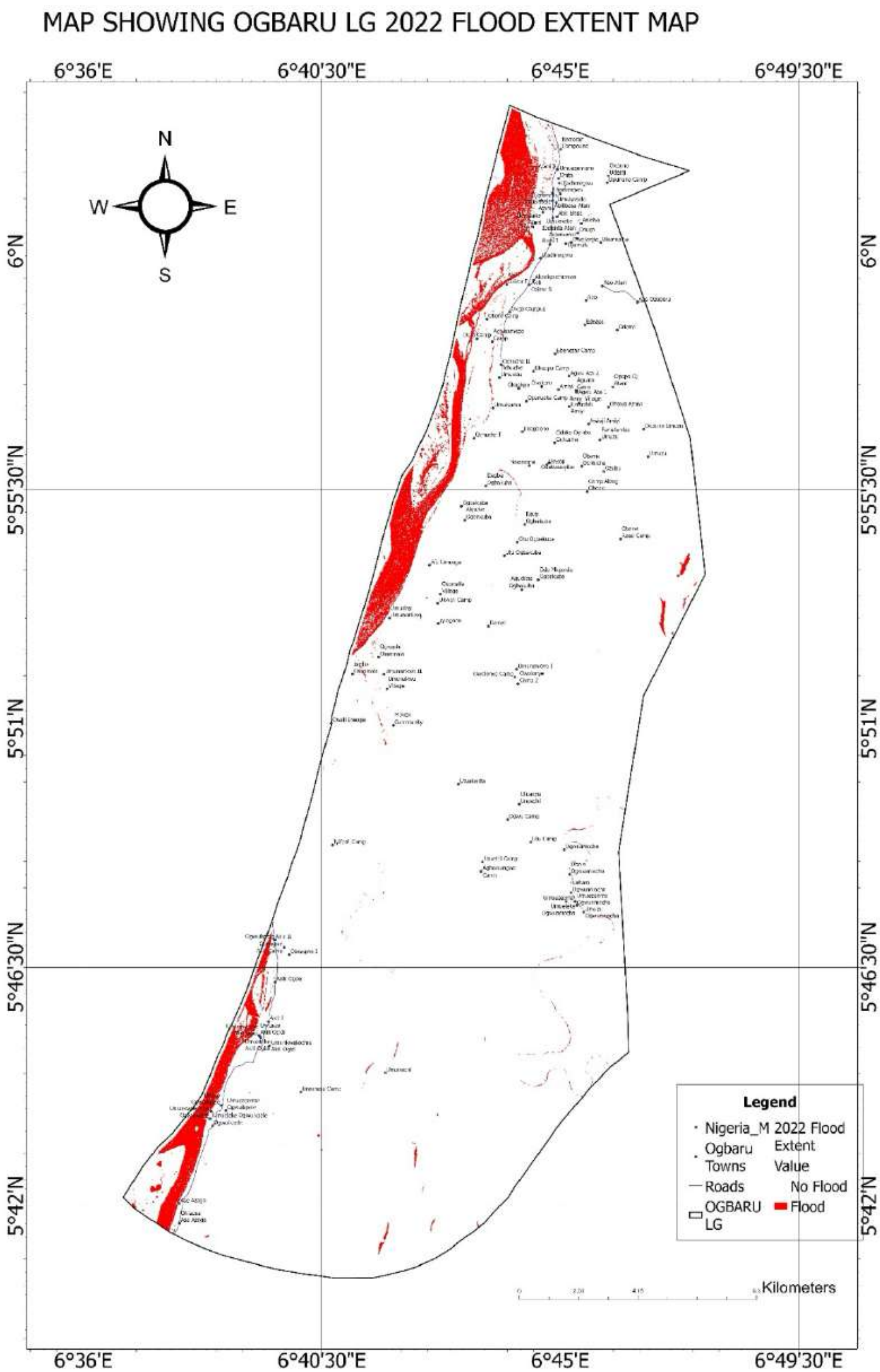

Figure 4.4: Flood extent map for 2022, highlighting peak extents affecting over 39% of vulnerable lands.

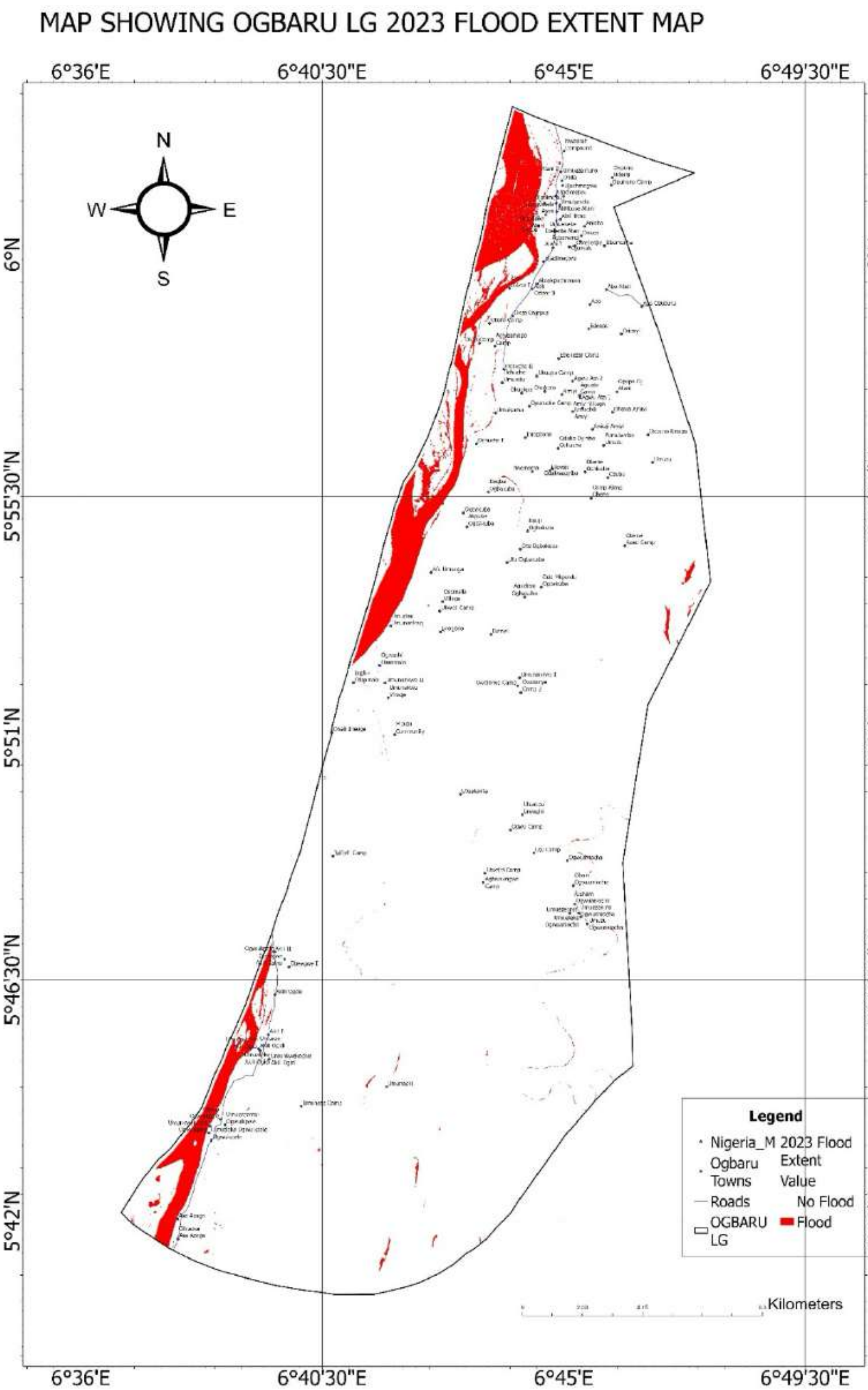

Figure 4.5: Flood extent map for 2023, indicating moderate recovery but persistent riverine threats.

These spatial patterns underscore topographic controls, with flatter slopes (<2°) amplifying spread.

4.2 LAND USE LAND COVER (LULC) DYNAMICS

4.2.1 What LULC Indicates

LULC classification via Random Forest on Sentinel-2 imagery categorizes surfaces into Water, Vegetation, Built-up, Bareland, and Forest. Before-and-after flood maps reveal shifts, such as vegetation loss to water post-flood, informing impact assessments.

4.2.2 Temporal and Pre/Post-Flood Shifts

Pre-flood LULC showed dominant vegetation (agriculture/forests ~50-60%), while post-flood maps indicated conversions to water/bareland. For instance, in 2018, vegetation decreased markedly post-flood due to submersion. Overall accuracy exceeded 85%, with Kappa >0.80.

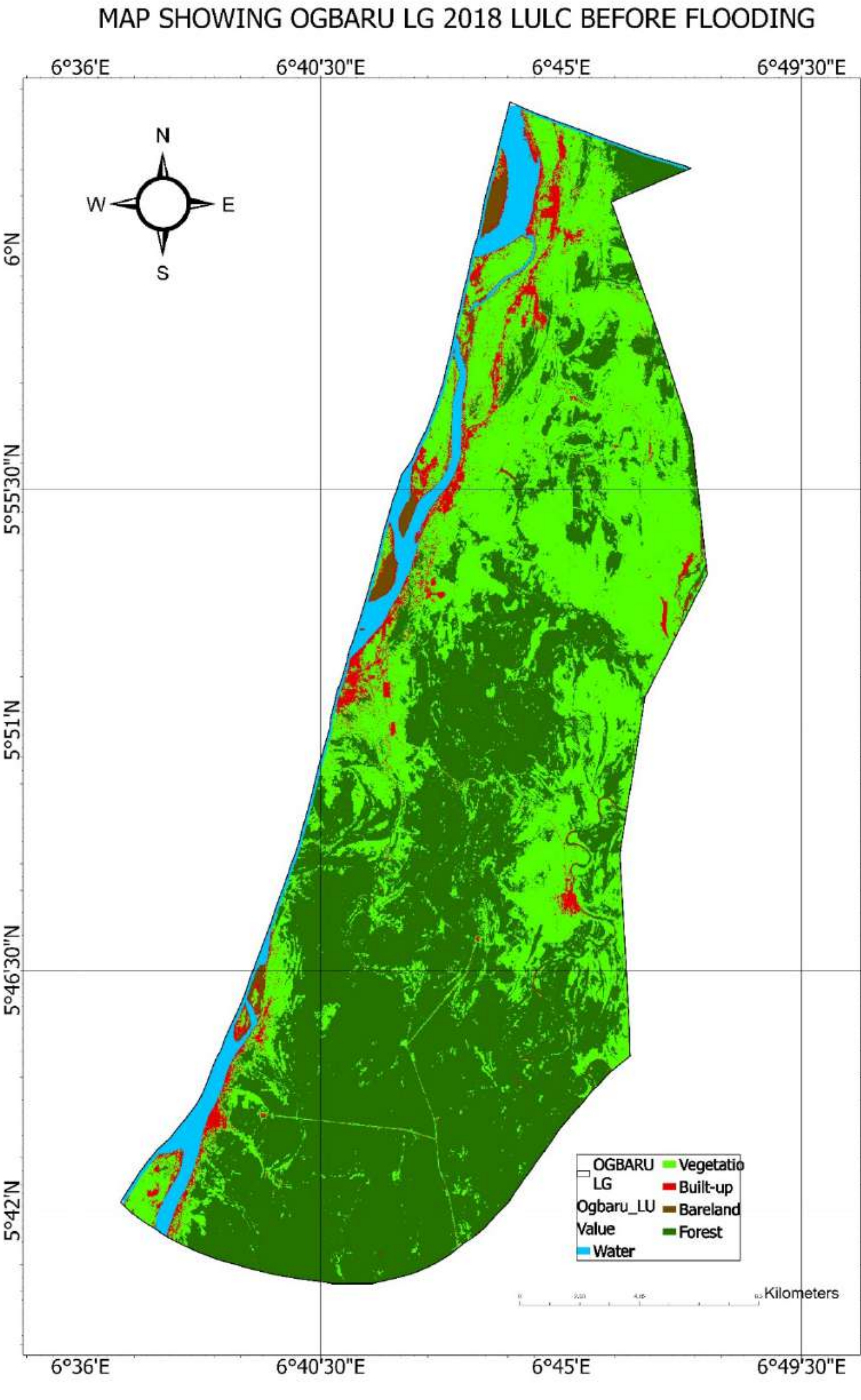

Figure 4.6: Pre-flood LULC map for 2018, dominated by green vegetation and yellow built-up areas.

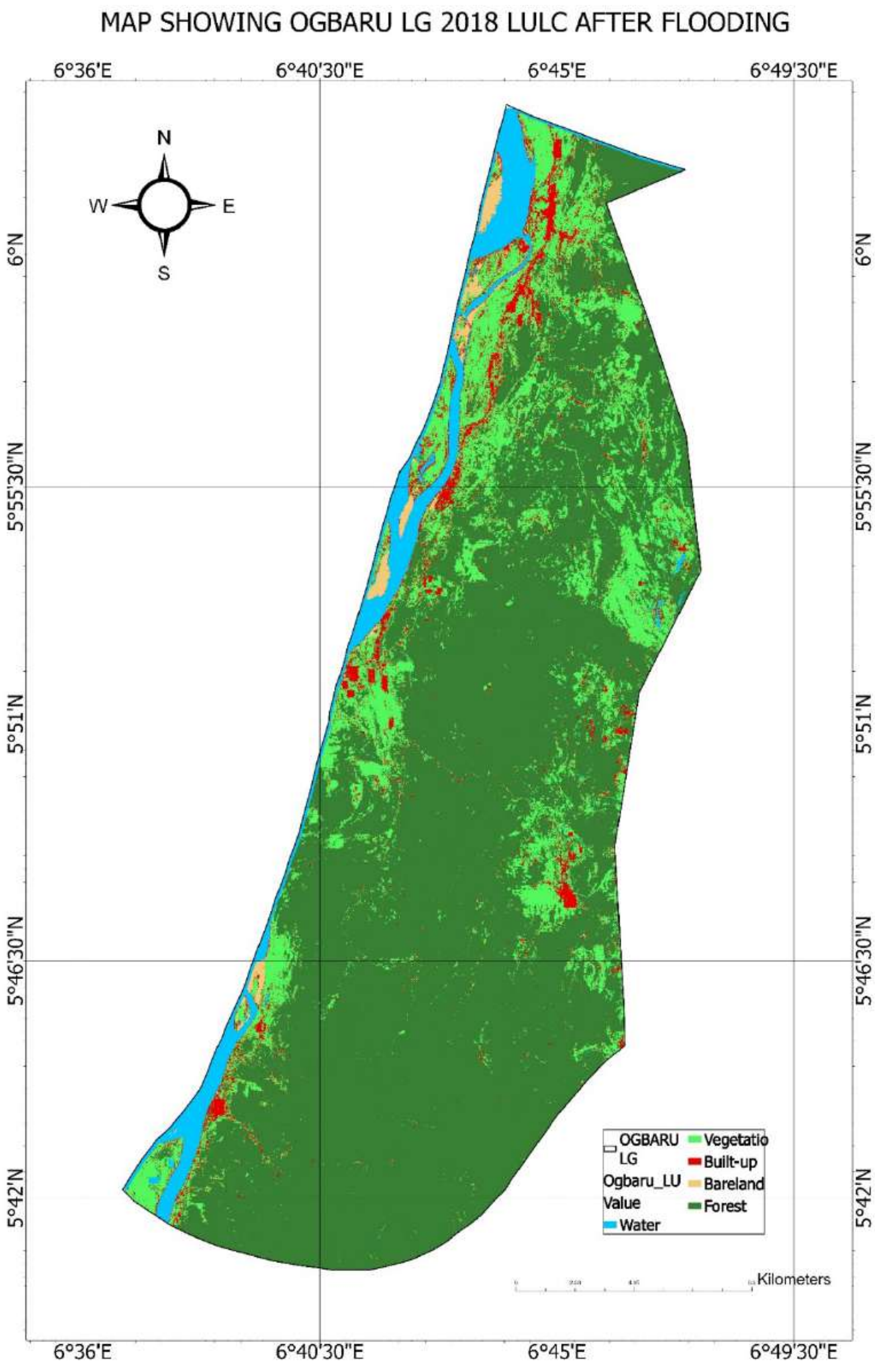

Figure 4.7: Post-flood LULC map for 2018, showing expanded blue water classes over former vegetation.

Similar patterns held for other years: 2020 exhibited minimal changes, while 2022 post-flood revealed fragmented forests. By 2023, recovery was evident, but cumulative vegetation loss reached ~20% over the period, driven by recurrent inundation and urban expansion.

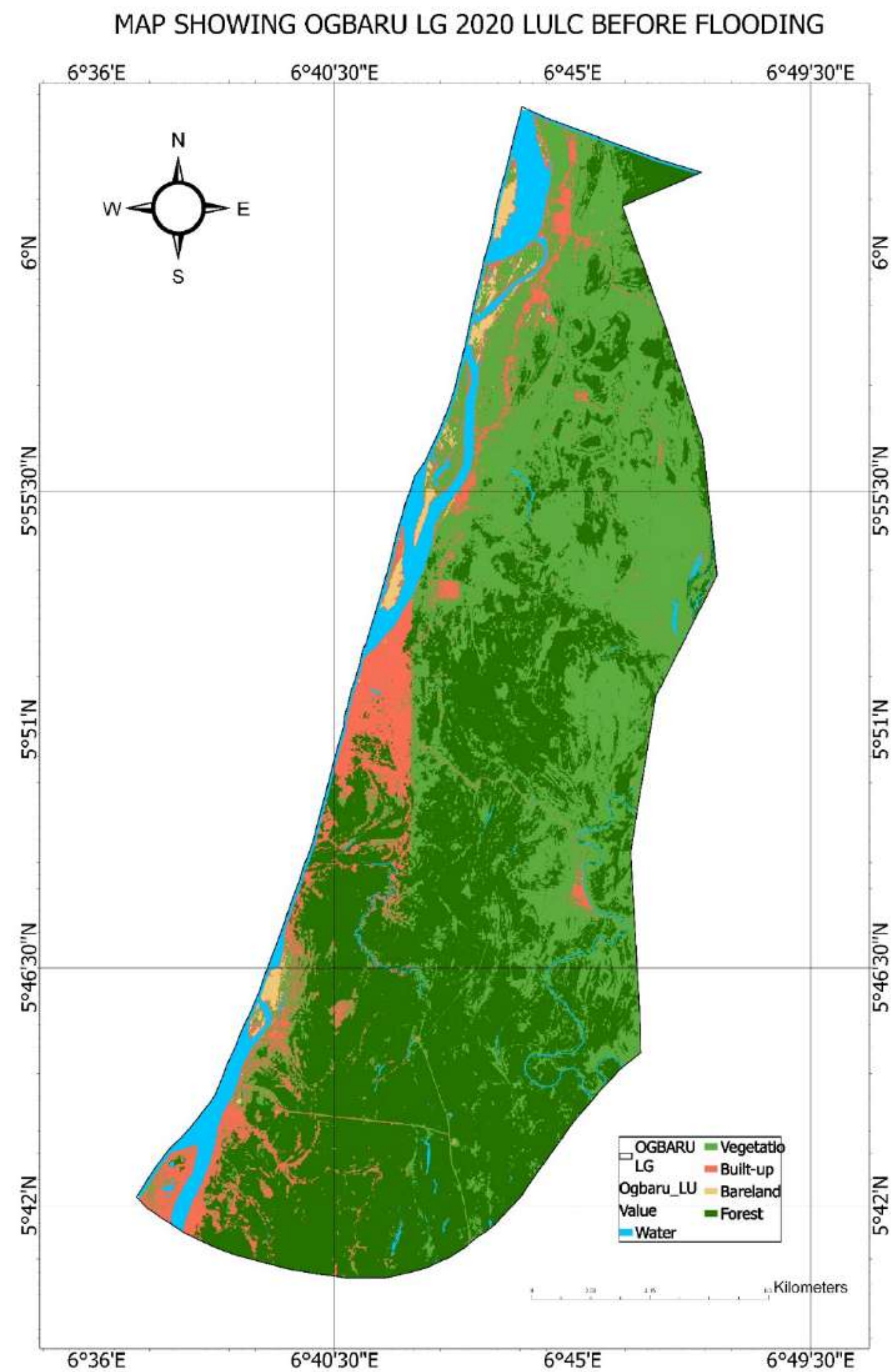

Figure 4.8: Pre-flood LULC map for 2020.

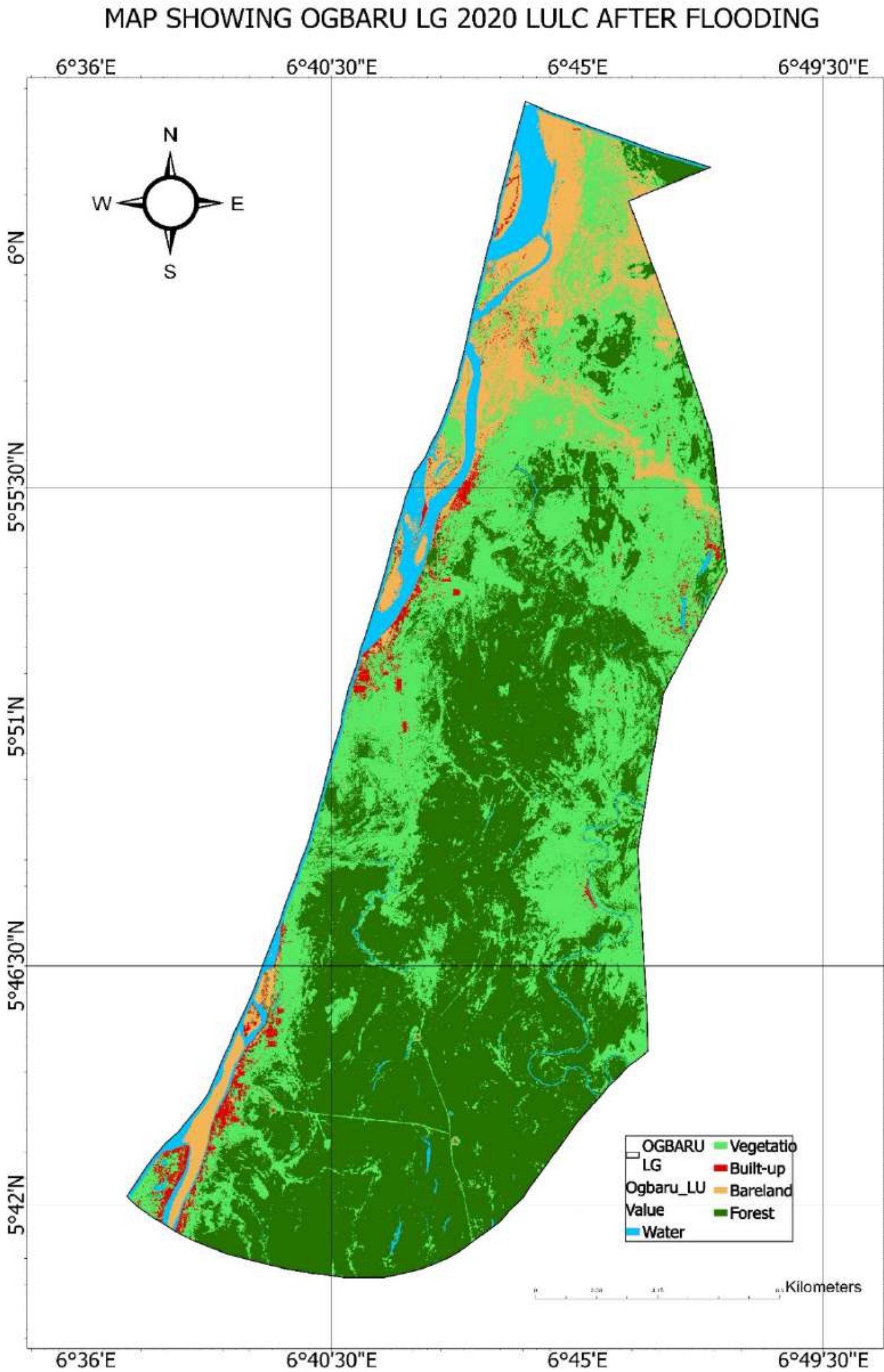

Figure 4.9: Post-flood LULC map for 2020.

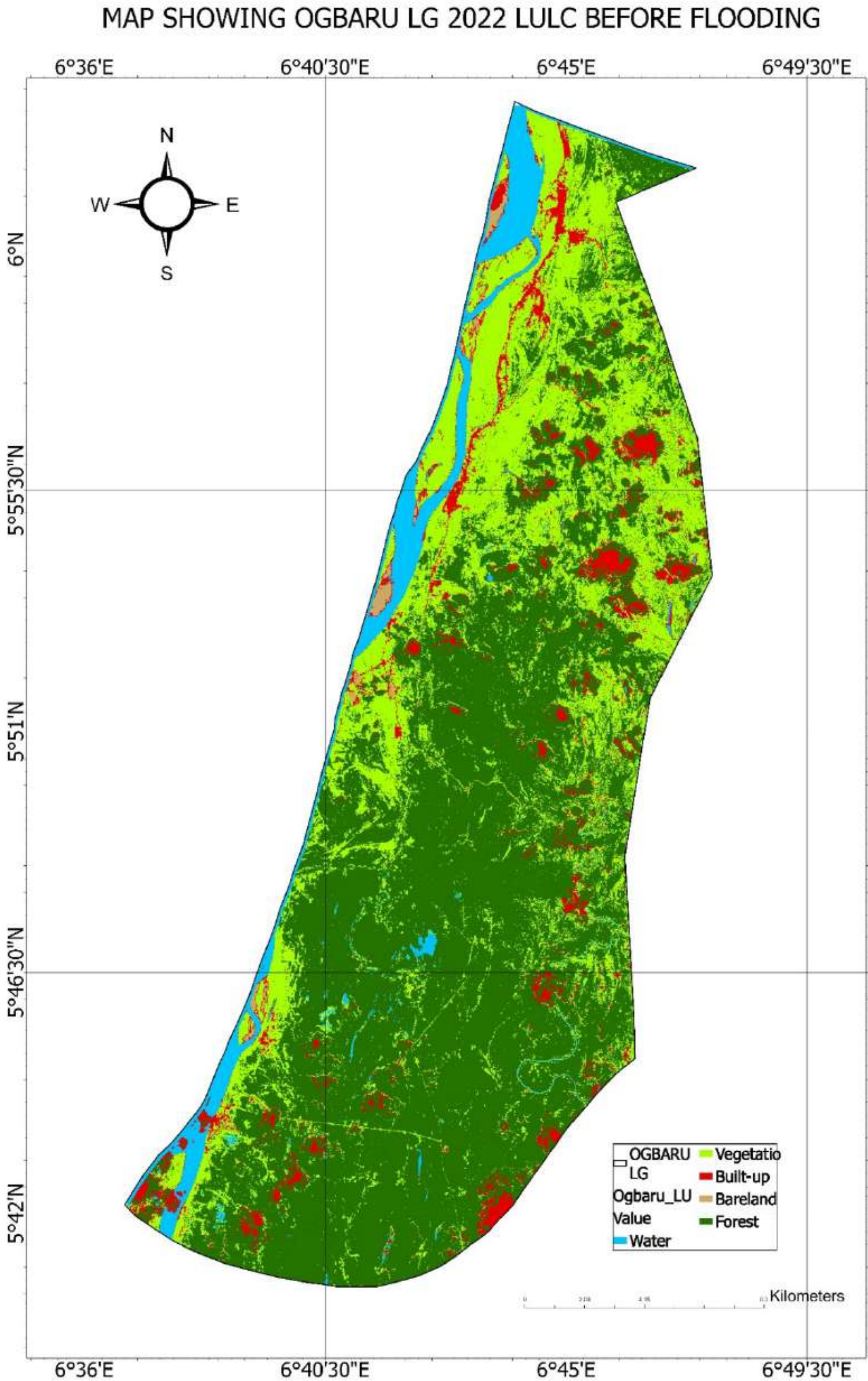

Figure 4.10: Pre-flood LULC map for 2022.

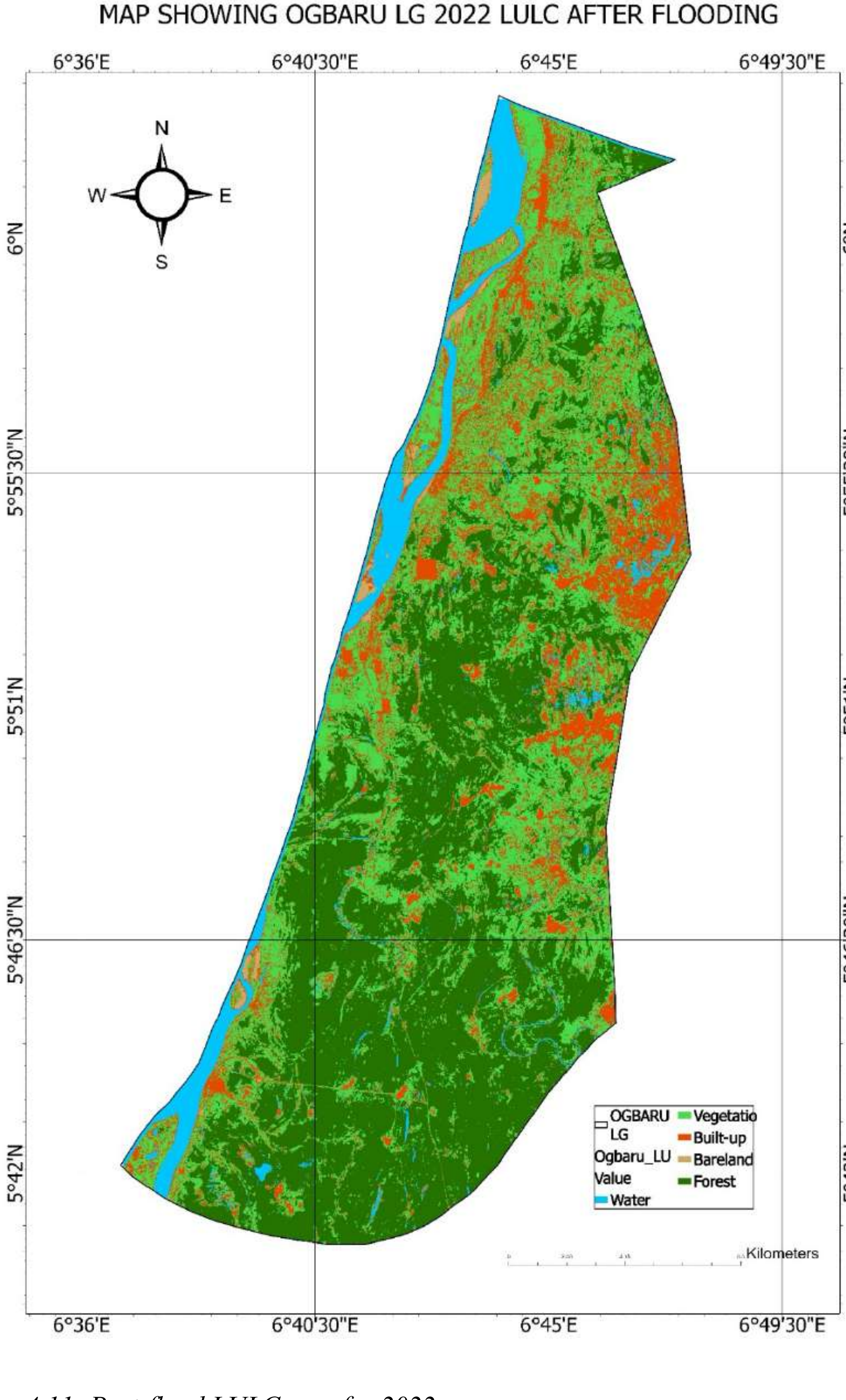

Figure 4.11: Post-flood LULC map for 2022.

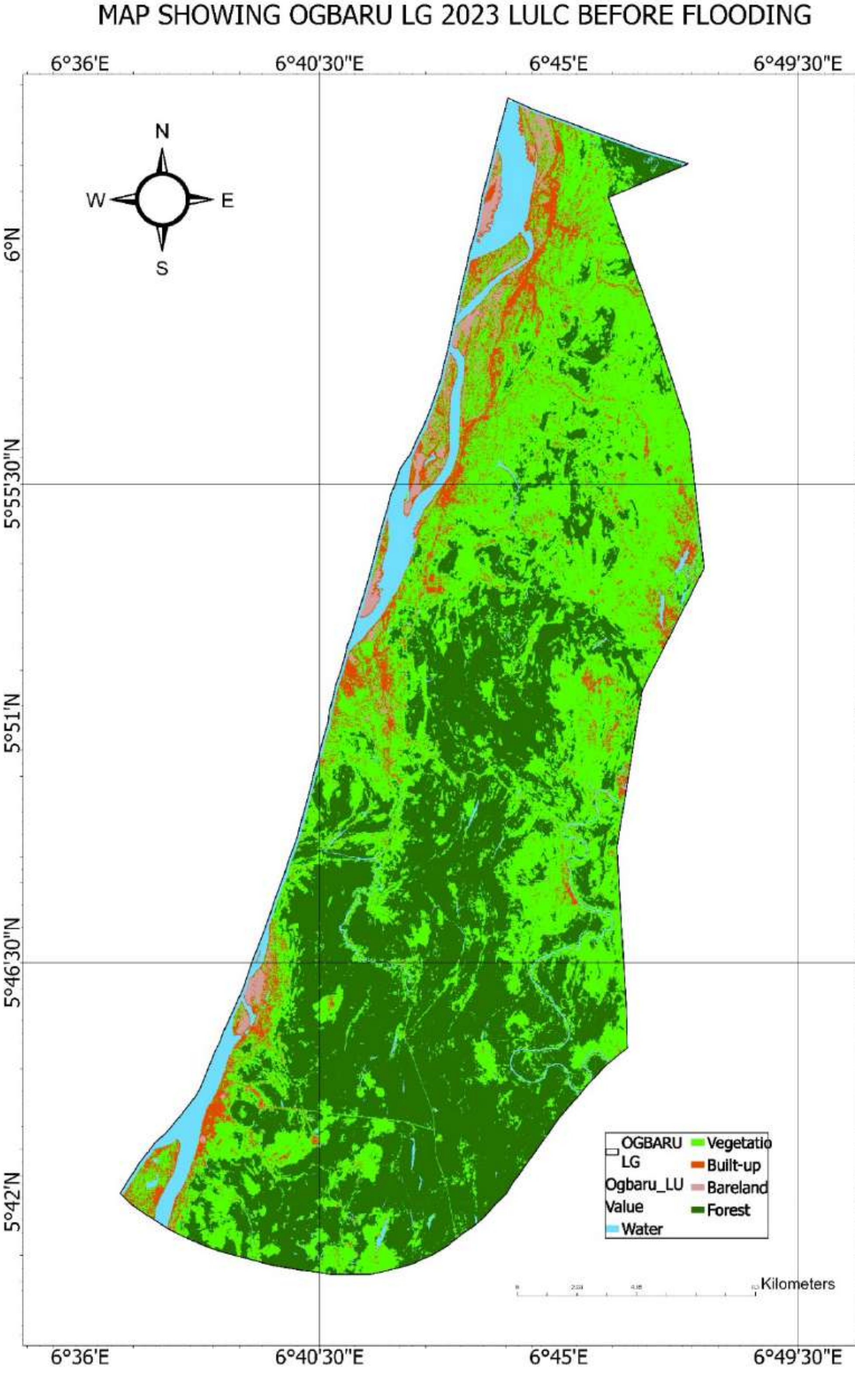

Figure 4.12: Pre-flood LULC map for 2023.

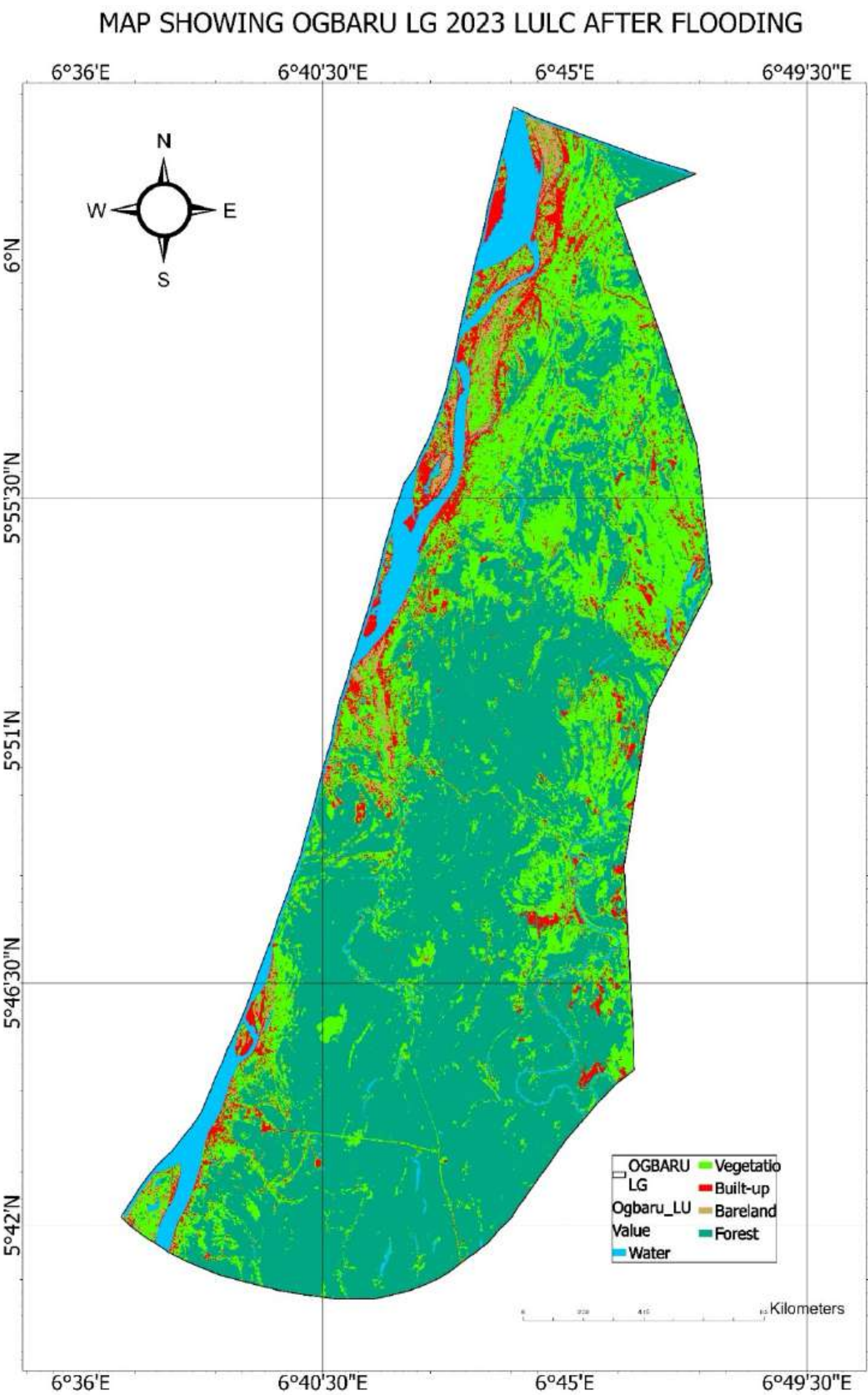

Figure 4.13: Post-flood LULC map for 2023.

These maps illustrate flood-induced degradation, with bareland increasing post-event due to erosion.

4.3 FLOOD IMPACT ANALYSIS

4.3.1 Overlay Results

Overlaying flood masks with LULC quantified impacts by class (ha):

Year	Water	Vegetation	Built-up	Bareland	Forest
2018	1416.622	419.4422	301.7187	291.9358	41.23586
2020	1582.177	104.022	131.2041	225.9021	3.390934
2022	1372.084	129.8626	169.7201	120.8625	25.99823
2023	1645.724	59.95065	181.6408	209.5929	1.625639

Table 4.1: Flooded areas (ha) by LULC class, showing vegetation and built-up as most affected.

Agricultural impacts (flooded vegetation/forest, proxy for farmland) peaked at 460.678 ha in 2018, dropping to 61.576 ha in 2023, reflecting variability but cumulative losses exceeding 750 ha.

4.3.2 Spatial and Sectoral Impacts

Impact maps highlight infrastructure risks: In 2018, ~300 ha built-up flooded, affecting settlements like Atani (roads submerged, facilities isolated). Agricultural losses were severe in Ossomala farmlands, aligning with NEMA's 2018 reports of displaced populations. By 2022, despite lower totals, built-up impacts rose, indicating urban encroachment into floodplains.

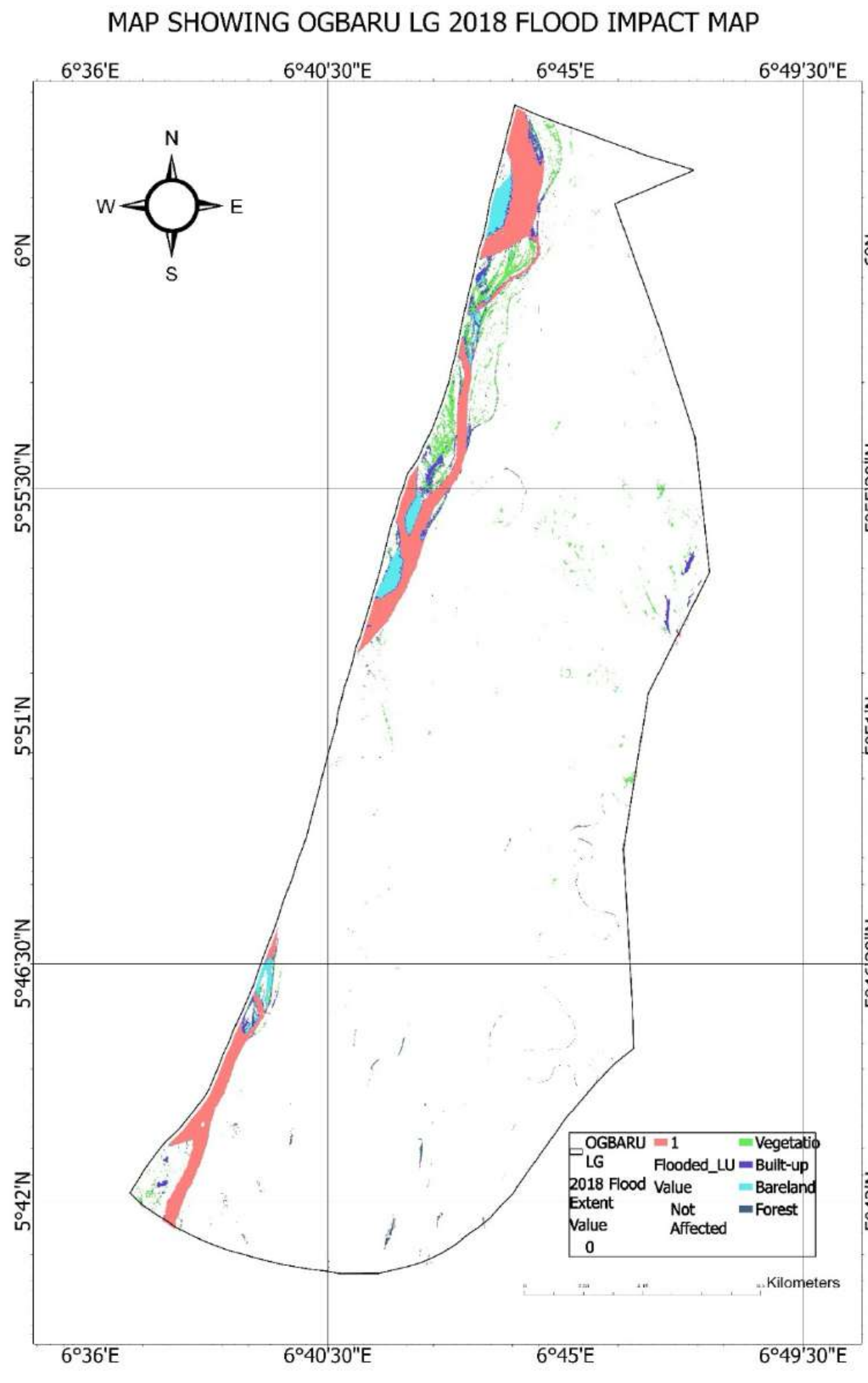

Figure 4.14: Flood impact map for 2018, overlaying extents on LULC with red hotspots for high-impact zones.

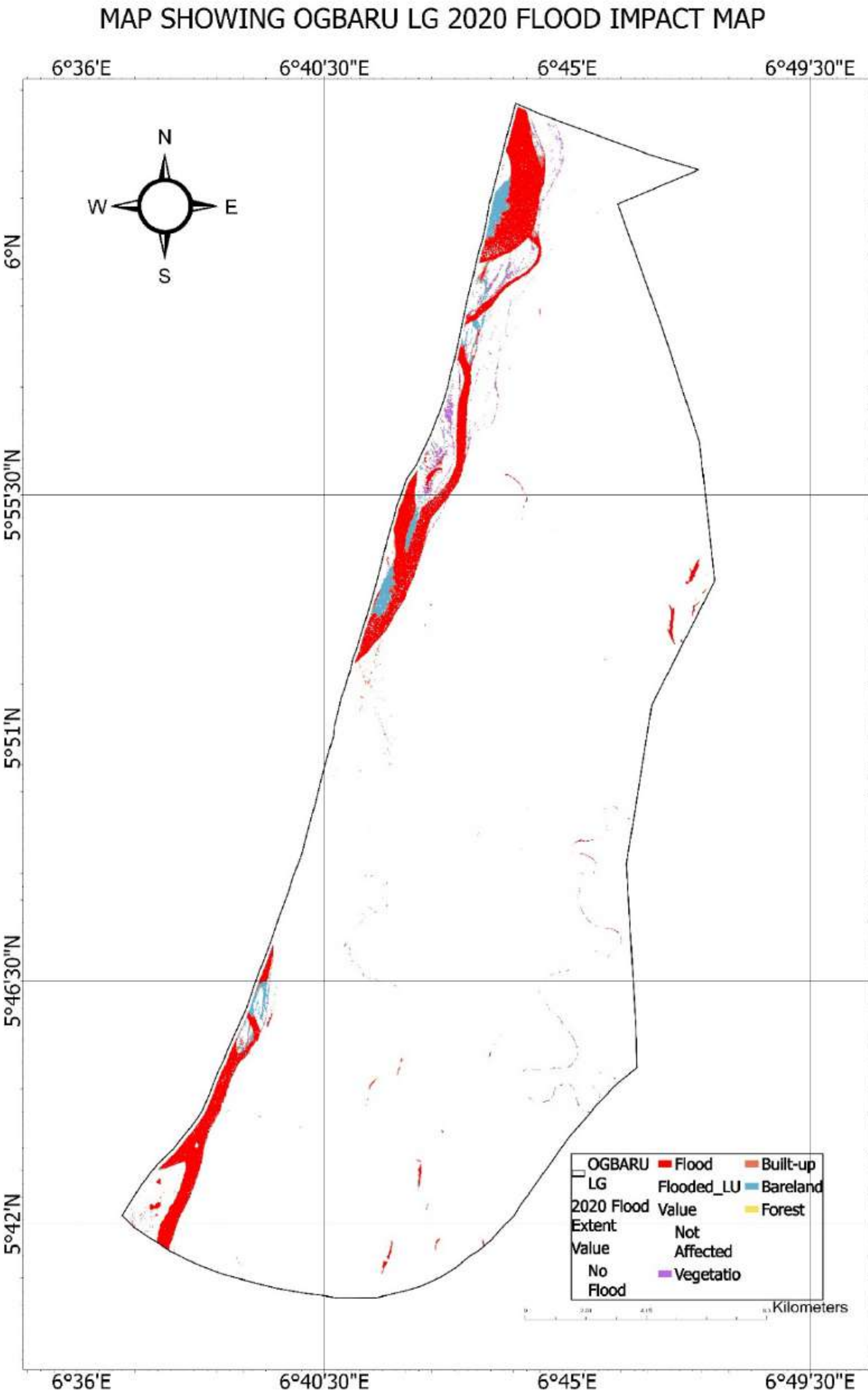

Figure 4.15: Flood impact map for 2020.

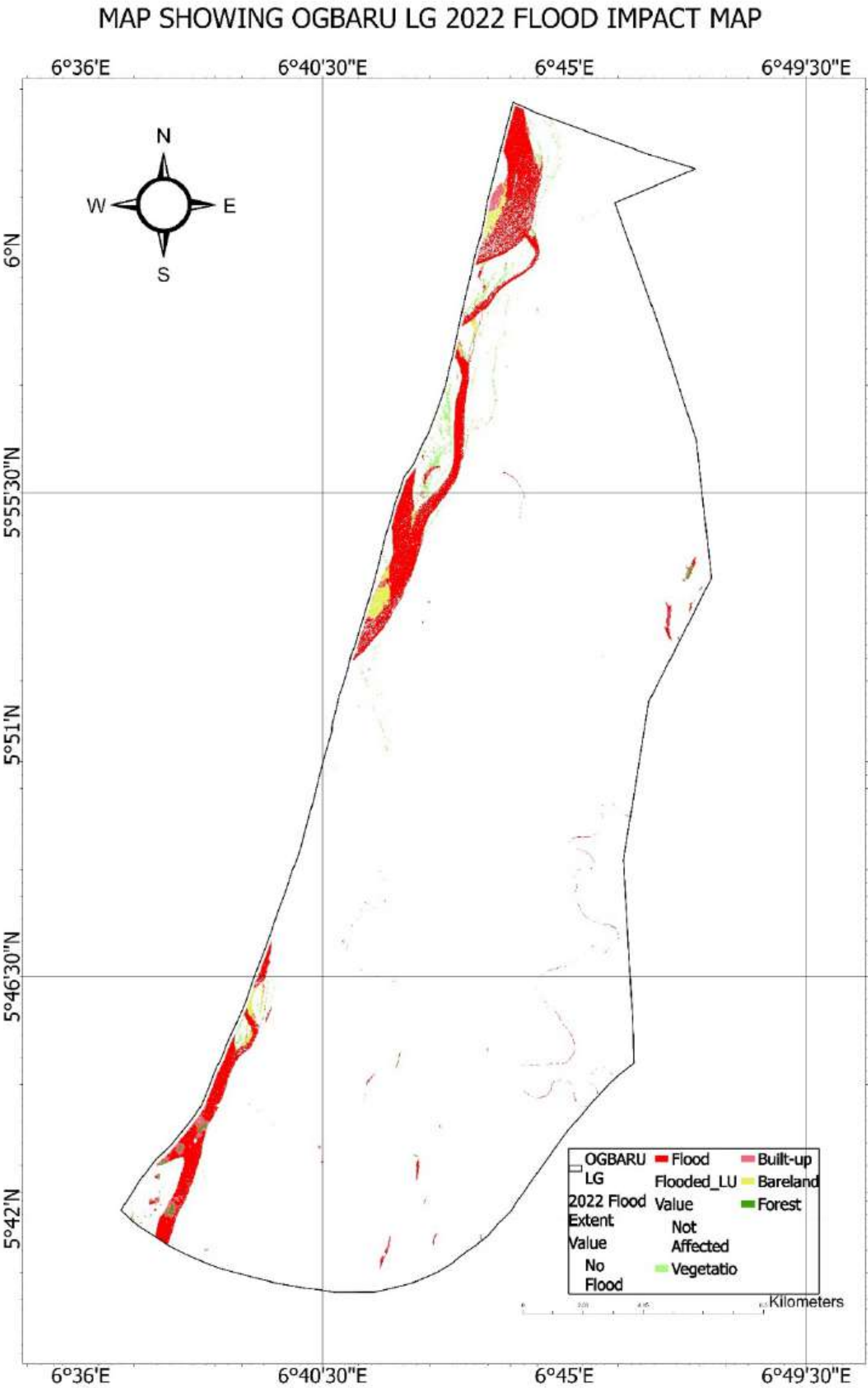

Figure 4.16: Flood impact map for 2022.

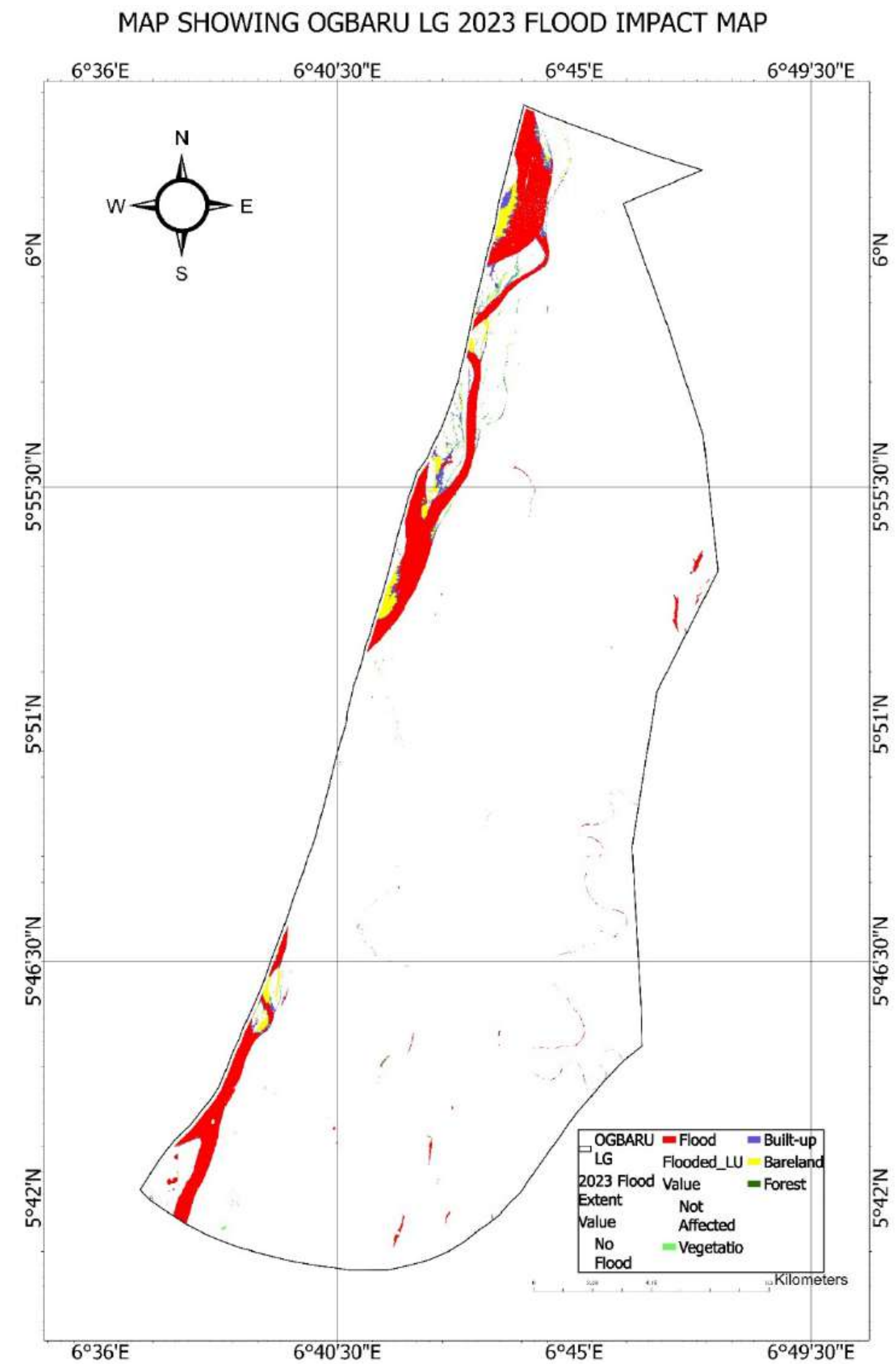

Figure 4.17: Flood impact map for 2023.

These visualizations reveal economic tolls, with agriculture (rice/yam fields) suffering most, potentially reducing yields by 30-50% per event.

4.4 VULNERABILITY ASSESSMENT

4.4.1 FVI Calculation and Findings

The FVI, derived from weighted overlays of elevation, slope, LULC, population, and flood history, classified zones as Low (green), Moderate (yellow), High (red). High vulnerability covered ~40% of Ogbaru, focused on densely populated riverine areas (>500/km²) with low elevation.

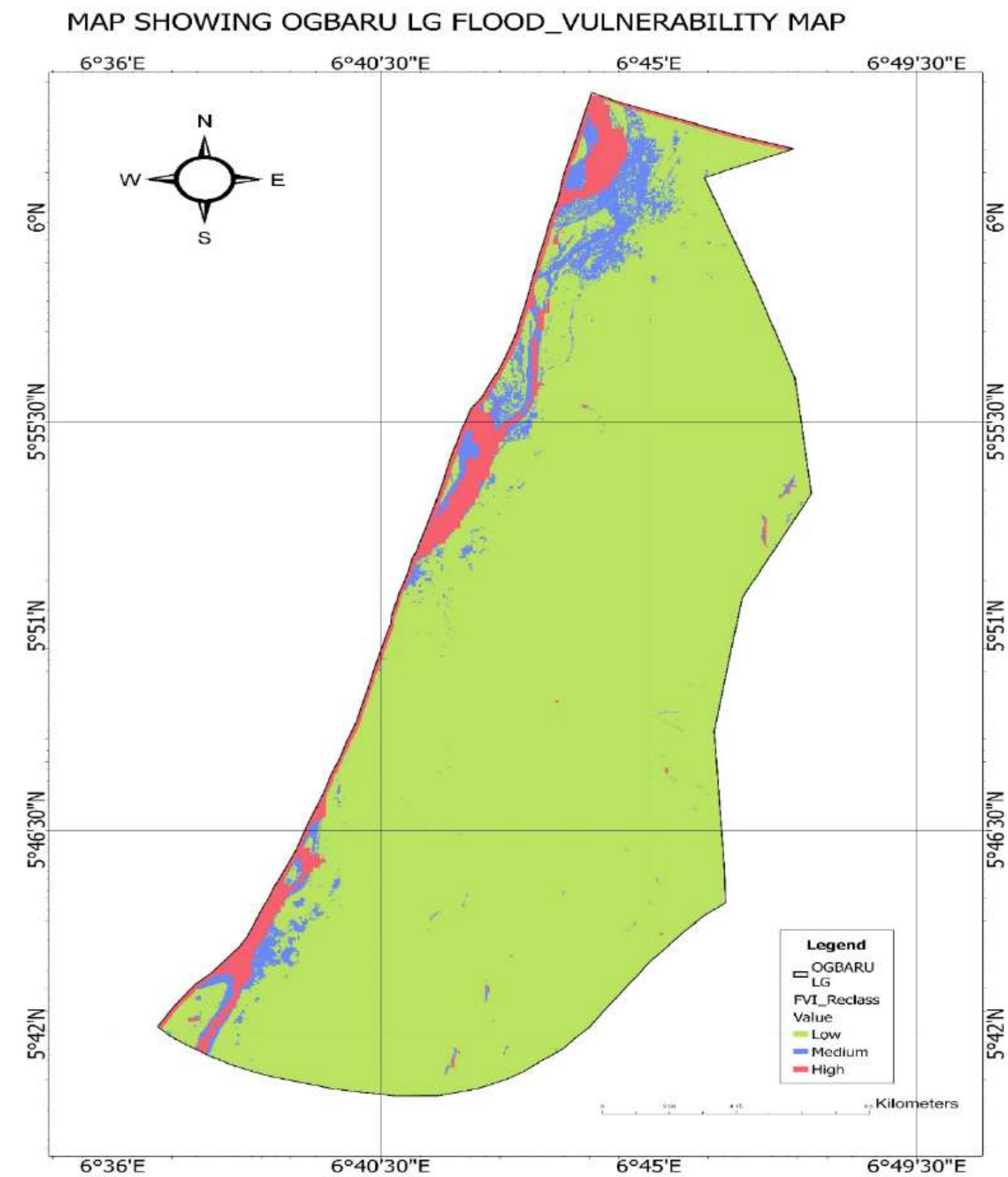

Figure 4.18: Flood vulnerability map, delineating risk zones based on multi-criteria analysis.

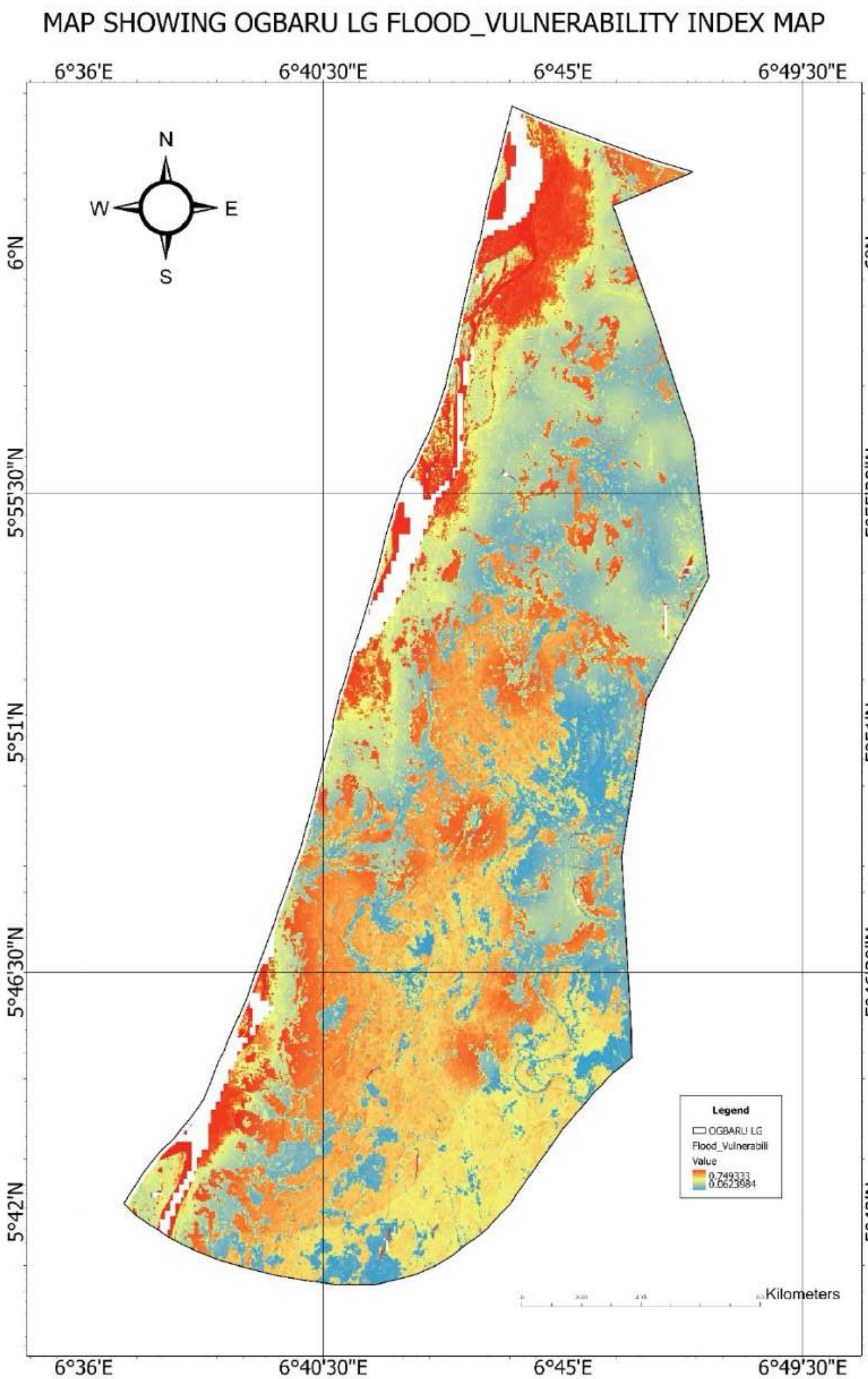

Figure 4.19: Detailed FVI map, with index values from 0-1, highlighting high-risk (>0.66) in red along rivers.

Spatial analysis shows 60% of population in high-vulnerability zones, exacerbating socio-economic risks.

4.5 IMPLICATIONS AND DISCUSSION

4.5.1 Environmental and Hydrological Insights

Flood trends correlate positively (r=0.65) with CHIRPS rainfall, underscoring climate variability's role. LULC shifts amplify runoff, as vegetation loss reduces infiltration. Compared to literature (e.g., Nigerian flood studies), Ogbaru's patterns mirror Niger Delta dynamics, where deforestation heightens erosion.

4.5.2 Socio-Economic Ramifications

Impacts disproportionately affect agriculture-dependent communities, with 2018 losses displacing thousands and contaminating water sources. Built-up flooding disrupts trade (e.g., naval base access), costing millions in damages. Vulnerability maps inform targeted interventions, like elevating structures in high-FVI areas.

4.5.3 Policy and Mitigation Recommendations

Findings advocate for integrated flood management: early warning systems, green infrastructure (mangrove restoration), and zoning laws restricting floodplain development. Community-based adaptation, informed by these maps, could reduce vulnerabilities by 20-30%.

4.5.4 Limitations and Future Research

While robust, results may underestimate micro-floods due to resolution. Future studies could incorporate in-situ data for validation.

4.6 SUMMARY OF FINDINGS

1. Flood extents averaged 2159 ha annually, peaking in 2018.
2. LULC showed post-flood vegetation declines, with 20% cumulative loss.
3. Agricultural impacts totaled over 750 ha, highest in 2018.
4. High vulnerability zones encompass 40% of Ogbaru, affecting dense populations.

5. Strong rainfall-flood correlations highlight climate drivers.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.0 CONCLUSION

The study has identified key findings from the geospatial assessment of flood impact and vulnerability in Ogbaru Local Government Area (LGA), Anambra State, Nigeria, covering the period from 2018 to 2023. The study utilized multi-temporal satellite remote sensing and GIS techniques to analyze spatial and temporal flood patterns, assess vulnerable areas, and evaluate the broader implications for disaster risk management and sustainable development. The results highlight critical insights that can inform local planning and policy-making. This chapter also addresses the limitations encountered during the study and suggests directions for future research. Furthermore, it reflects on how the research objectives outlined in Chapter One were achieved through the methodology detailed in Chapter Three and the results presented in Chapter Four.

5.1 SUMMARY OF KEY FINDINGS

The geospatial assessment successfully achieved its primary aim of providing a detailed understanding of flood dynamics, impacts, and vulnerability in Ogbaru LGA. The key findings are summarized as follows:

1. **Flood Extent and Temporal Dynamics:** The study revealed that Ogbaru LGA experienced significant annual inundation, with an average flooded area of 2159 hectares over the 2018-2023 period, representing approximately 5.6% of the LGA's total landmass. Flood extents peaked notably in 2018, recording 2470.954 hectares, followed by variations in subsequent years, including a rebound in 2023 (2098.534 ha). This inter-annual variability of flood extents showed a strong positive correlation (r=0.65) with CHIRPS rainfall data, underscoring the direct influence of climate variability on flood occurrences in the region. Spatially, inundation was consistently concentrated along the low-elevation (below 20 meters) floodplains of the River Niger and Orashi River, confirming these areas as perennial flood hotspots.

2. **Land Use/Land Cover (LULC) Dynamics:** The analysis of LULC revealed significant transformations, particularly between pre- and post-flood periods. Dominated by vegetation (agriculture and forests) in pre-flood conditions, considerable conversion to water and bareland classes was observed immediately following major flood events. Over the entire study period, Ogbaru LGA experienced a cumulative vegetation loss of approximately 20%, indicating persistent environmental degradation likely driven by recurrent inundation and, to some extent, unchecked urban expansion into vulnerable areas. The high accuracy (>85%}) and Kappa coefficients (>0.80) of the Random Forest classification validated the reliability of these LULC change detections.

3. **Flood Impact Assessment:** Overlay analyses quantified the tangible impacts of floods across different LULC classes. Agricultural lands, represented by vegetation and forest cover, bore the brunt of the impact, with cumulative losses exceeding 750 hectares over the five-year period, peaking at 460.678 hectares in 2018. Built-up areas were also significantly affected, with approximately 300 hectares flooded in 2018. While overall flood extents fluctuated, the impact on built-up areas demonstrated a concerning trend of increasing vulnerability in urbanized settlements like Atani, Odekpe, Ossomala, and Okpoko, suggesting human encroachment into high-risk zones. These findings are consistent with NEMA's reports of widespread displacement and livelihood disruptions.

4. **Flood Vulnerability Assessment:** The study successfully generated a comprehensive Flood Vulnerability Map for Ogbaru LGA by applying a multi-criteria approach, specifically an Analytic Hierarchy Process (AHP)-inspired weighting for the Flood Vulnerability Index (FVI). The assessment revealed that approximately 40% of Ogbaru LGA is characterized by high vulnerability to floods. These highly vulnerable zones are predominantly located in the low-elevation, densely populated riverine areas, where population densities often exceed 500 inhabitants per square kilometer. Crucially, the spatial analysis further revealed that about 60% of Ogbaru’s total population resides within these identified high-vulnerability zones, significantly exacerbating the socio-economic risks associated with flood occurrences. The weighting scheme (Elevation 30%, Flood History 25%, Population 20%, LULC 15%, Slope 10%) highlighted the dominant role of topography and past flood events in determining current vulnerability.

5.2 DISCUSSION OF IMPLICATIONS

The findings of this study carry profound implications for disaster risk reduction, land use planning, and climate change adaptation strategies within Ogbaru LGA and similar riverine communities across Nigeria.

Firstly, the quantitative mapping of flood extents and their strong correlation with rainfall patterns provide empirical evidence necessary for developing and refining early warning systems. Knowing *where* floods consistently occur and *how much* area is affected annually offers critical baseline data for more accurate flood forecasting and timely dissemination of alerts to vulnerable communities.

Secondly, the detailed LULC change analysis, particularly the observed vegetation loss and increasing built-up area impacts, underscores the urgent need for sustainable land management practices. Unregulated urban expansion into floodplains not only puts more assets at risk but also degrades natural flood absorption capacities, creating a vicious cycle of increasing vulnerability. This implies that current land use policies, if they exist, are either inadequate or poorly enforced.

Thirdly, the comprehensive flood impact assessment, highlighting specific agricultural and infrastructural losses, provides a strong economic justification for investment in flood mitigation. The cumulative loss of over 750 hectares of agricultural land directly threatens food security and the livelihoods of an agrarian population, necessitating resilient agricultural practices and diversification strategies. The rising impacts on built-up areas further emphasize the economic tolls and the need for flood-resistant infrastructure development.

Finally, the meticulously generated Flood Vulnerability Map is an invaluable planning tool. By spatially delineating high-risk zones and identifying the density of populations within them, the map enables targeted interventions. Policy-makers and disaster management agencies can prioritize resource allocation, initiate community-based adaptation programs (e.g., promoting elevated housing, relocating critical facilities), and enforce strict zoning regulations to prevent further development in high-FVI areas. This shift from reactive response to proactive risk reduction, informed by geospatial intelligence, is critical for building long-term resilience.

5.3 RECOMMENDATIONS

Based on the comprehensive geospatial assessment of flood impact and vulnerability in Ogbaru LGA (2018-2023), the following recommendations are put forth to enhance flood risk management and foster community resilience:

1. **Strengthen Early Warning Systems (EWS):** Leverage the positive correlation between rainfall and flood extents by investing in real-time hydrological monitoring stations along the River Niger and its tributaries in Ogbaru. Integrate this data with satellite-derived flood maps (like those generated in this study) to provide accurate, timely, and localized flood forecasts and alerts to communities, potentially reducing vulnerabilities by 20-30%.
2. **Implement Integrated Land Use Planning and Zoning Laws:** Enforce strict zoning regulations that restrict new urban development, agricultural expansion, and other human activities within identified high-vulnerability floodplains (i.e., the approximately 40% of Ogbaru classified as highly vulnerable). Promote flood-compatible land uses, and consider managed retreat or elevated construction for existing structures in chronically inundated areas.
3. **Promote Green Infrastructure and Ecosystem-Based Adaptation (EbA):** Invest in the restoration and conservation of natural flood-mitigating ecosystems, specifically mangrove and wetland restoration along the riverine areas of Ogbaru. These natural buffers can significantly enhance water retention, reduce runoff velocity, and mitigate erosion, thereby compensating for the observed 20% cumulative vegetation loss.
4. **Support Resilient Agriculture and Livelihood Diversification:** Introduce and support flood-resilient agricultural practices (e.g., cultivation of flood-tolerant crop varieties, raised bed farming) in affected communities. Additionally, promote livelihood diversification away from solely flood-dependent sectors to reduce economic vulnerability stemming from recurrent agricultural losses (over 750 ha in the study period).
5. **Enhance Community-Based Adaptation (CBA):** Utilize the flood vulnerability maps to engage local communities in high-FVI zones in participatory planning for adaptation strategies. This includes community-led initiatives for clearing drainage channels, identifying safe evacuation routes, building elevated communal shelters, and fostering local preparedness teams.

6. **Invest in Critical Infrastructure Resilience:** Prioritize flood-proofing and elevating essential infrastructure, particularly roads, schools, and health facilities, in the identified high-impact zones like Atani, Odekpe, and Ossomala. This will minimize disruption to essential services during flood events.

5.4 Limitations of the Study

Despite the robustness of the methodology and the significance of the findings, this study acknowledges certain limitations:

1. **Data Resolution:** While Sentinel-1 and Sentinel-2 provided 10-meter resolution data, this may still have overlooked micro-scale floods or very narrow infrastructure elements, potentially underestimating the total impact in highly localized areas.
2. **SAR Artifacts:** Although pre-processing techniques were applied, SAR imagery is susceptible to artifacts such as wind-roughened water surfaces or dense vegetated areas that can be misinterpreted as dry land, leading to potential misclassification (false negatives) of inundated areas.
3. **Cloud Interference (Optical Data):** Despite the reliance on SAR for flood mapping, persistent cloud cover during the wet season still limited the availability of perfectly cloud-free Sentinel-2 imagery for post-flood LULC accuracy assessment in all instances, potentially introducing some uncertainty in very localized areas.
4. **Static Ancillary Data:** Ancillary datasets like the Digital Elevation Model (DEM) and population estimates (WorldPop/HRSL) were static for the study period. While generally stable, dynamic changes in elevation due to erosion/sedimentation or annual population shifts could not be captured, potentially leading to slight underestimations of dynamic vulnerability.
5. **Validation Constraints:** Due to logistical and accessibility challenges in remote parts of Ogbaru LGA, ground-truth data for direct validation was scarce. The study relied on a combination of visual interpretation, cross-validation with cloud-free optical imagery, and consistency with media reports for accuracy assessment.
6. **Weighting Subjectivity (FVI):** While the Analytic Hierarchy Process (AHP)-inspired weighting for the Flood Vulnerability Index (FVI) was based on established literature and

expert judgment, the assigned weights can inherently be sensitive to adjustments and subjective interpretations.

5.5 Suggestions for Future Research

Based on the limitations identified and new insights gained, the following areas are recommended for future research:

1. **Integration of In-situ Data:** Future studies should aim to incorporate more extensive in-situ (ground-truth) data collection for enhanced validation of flood extent maps, LULC classifications, and vulnerability assessments, particularly in remote areas of Ogbaru LGA.
2. **Higher Resolution Data and Micro-Flood Analysis:** Exploring the use of very high-resolution satellite imagery (e.g., from commercial satellites if accessible) could enable the mapping of micro-scale floods and detailed assessment of impacts on small infrastructure, providing more granular insights.
3. **Dynamic Vulnerability Modeling:** Future research could focus on developing dynamic flood vulnerability models that integrate annually updated ancillary data (e.g., population redistribution models, annual land subsidence data if available) to capture more nuanced temporal shifts in vulnerability.
4. **Socio-Cultural Dimensions of Vulnerability:** Supplementing geospatial analyses with qualitative research methods (e.g., household surveys, focus group discussions) could provide deeper insights into socio-cultural vulnerabilities, coping capacities, and local perceptions of flood risk, enriching the understanding of human dimensions of vulnerability.
5. **Hydrodynamic Flood Modeling:** Incorporating hydrodynamic flood modeling techniques (e.g., using HEC-RAS or other models) could allow for the simulation of different flood scenarios (e.g., 1-in-50-year flood, 1-in-100-year flood), providing predictive capabilities for future flood events and assessing the effectiveness of proposed mitigation measures.
6. **Economic Cost-Benefit Analysis:** Future studies could quantify the economic costs of flood impacts and the benefits of proposed mitigation strategies in monetary terms, providing a more robust basis for investment decisions.

In conclusion, this geospatial assessment has provided a critical, data-driven understanding of flood impact and vulnerability in Ogbaru LGA. The findings serve as a vital resource for proactive disaster risk reduction, advocating for integrated approaches that combine advanced geospatial technologies with informed policy and community engagement to build a more resilient future for the affected populations.

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